

Introduction

The STM32 Motor Control Software Development Kit (MC SDK), is part of the STMicroelectronics motor-control ecosystem. It includes:

- ST MC FOC FW library for Permanent-Magnet Synchronous Motor (PMSM) field-oriented control (FOC)
- ST MC 6 Step FW Library
- ST Motor Profiler
- ST Motor Pilot
- ST MC Workbench software tool, a Graphical User Interface (GUI) for the configuration of MC SDK firmware library parameters

This User's Manual explains How To use the ST MC Workbench software tool included with this MC SDK.

Contents

Introduction	1
List of figures	4
1. General information	7
2. Related documents	8
3. The ST Motor Control Workbench.....	9
3.1 Editing field behavior.....	10
3.2 Hardware Cards	11
3.2.1 Cards Information	11
3.2.2 Cards Actions.....	11
3.2.3 Hardware Preferred	12
3.3 Project Composer Page	13
3.3.1 Composer Start View	13
3.3.2 Motor	13
3.3.3 Board	18
3.3.4 Composer Flow.....	21
3.3.5 Pack	26
3.4 Summary Page	27
3.4.1 Summary Header	27
3.4.2 Project's Available Features	28
3.4.3 Summary Schema.....	29
3.4.4 Summary Footer.....	30
3.5 Loading an existing Project.....	31
3.6 Project view	32
3.6.3 Toolbar area	33
3.6.4 Hardware information area	35
3.6.5 Project steps and hardware configuration details.....	37
3.7 Project wizard	38
3.7.3 Project Hw & Info	38
3.7.4 Motor	40
3.7.5 Power Supply.....	42
3.7.6 PWM Generation	42

3.7.7	Speed Sensing Selection.....	45
3.7.8	Current Sensing	46
3.7.9	Bus Voltage Sensing	56
3.7.10	Temperature Sensing.....	57
3.7.11	Speed Sensing Config.....	58
3.7.12	Driver Protection	67
3.7.13	Drive Settings	68
3.7.14	Stage Configuration	74
3.7.15	User Interface	75
3.7.16	Application Configuration.....	76
3.7.17	Pin usage & Hardware Changes.....	77

List of figures

Figure 1: ST MC Workbench – Icon	9
Figure 2: ST MC Workbench – GUI (Home View).....	9
Figure 3: ST MC Workbench – Editing field	10
Figure 4: ST MC Workbench – Unit and Legend Area.....	10
Figure 5: ST MC Workbench – Cards (Motor-Control-Power-Inverter-Adapter)	11
Figure 6: ST MC Workbench – Cards Actions.....	12
Figure 7: ST MC Workbench – Cards – Delete Custom Hardware Warning	12
Figure 8: ST MC Workbench – Composer window	13
Figure 9: ST MC Workbench – New Project window – Motors	14
Figure 10: ST MC Workbench – New Project – Motors – Filters	14
Figure 11: ST MC Workbench – New Project Window – Selected Motor	15
Figure 12: ST MC Workbench – New Project window – Dual Motor selection	16
Figure 13: ST MC Workbench – New Project window – Clone/Save as Motor	17
Figure 14: ST MC Workbench – New Project window – Save as Motor Error.....	17
Figure 15: ST MC Workbench – New Project Window – Board Selection	18
Figure 16: ST MC Workbench – New Project Window – Board Selection Power Filters.....	19
Figure 17: ST MC Workbench – New Project Window – Board Selection Control Filters	20
Figure 18: ST MC Workbench – New Project Window – Board Selection Inverter Filters	20
Figure 19: ST MC Workbench – New Project Window – Board Selection Cases	22
Figure 20: ST MC Workbench – New Project Window – Board Selection Need Adapter	23
Figure 21: ST MC Workbench – New Project Window – Pop-up for Adapter required	23
Figure 22: ST MC Workbench – New Project Window – Adapter Selection dialog.....	24
Figure 23: ST MC Workbench – New Project Window – Board Connections.....	25
Figure 24: ST MC Workbench – New Project Window – Pack Selection Dialog	26
Figure 25: ST MC Workbench – New Project Window – Summary Board Selection and Connections.....	27
Figure 26: ST MC Workbench – New Project Window – Summary Header	27
Figure 27: ST MC Workbench – New Project Window – Summary Algorithm not available	28
Figure 28: ST MC Workbench – New Project Window – Summary Available features	28
Figure 29: ST MC Workbench – New Project Window – Summary Warning Report	29
Figure 30: ST MC Workbench – New Project Window – Summary Stage motor	29
Figure 31: ST MC Workbench – New Project Window – Summary Schema	29
Figure 32: ST MC Workbench – Project view (global view)	32
Figure 33: ST MC Workbench – Close Project confirmation window	33
Figure 34: ST MC Workbench – Save menu	33
Figure 35: ST MC Workbench – Project Save As window	34
Figure 36: ST MC Workbench – Save & Generate	34
Figure 37: ST MC Workbench – Generate – Hardware Changes	34
Figure 38: ST MC Workbench – Project generation	35
Figure 39: ST MC Workbench – About.....	35
Figure 40: ST MC Workbench – MCU.....	36
Figure 41: ST MC Workbench – Control Board	36

Figure 42: ST MC Workbench – Power Board.....	37
Figure 43: ST MC Workbench – Project Hw & Info	39
Figure 44: ST MC Workbench – Project Hw & Info - Change motor.....	40
Figure 45: ST MC Workbench – Motor	41
Figure 46: ST MC Workbench – Motor – Hall Effect – Quadrature Encoder	42
Figure 47: ST MC Workbench – Power Supply	42
Figure 48: ST MC Workbench - PWM Generation - High and Low Side	43
Figure 49: ST MC Workbench - PWM Generation – High Side and One Enable.....	44
Figure 50: ST MC Workbench – PWM Generation – High Side and Three Enables.....	45
Figure 51: ST MC Workbench – Speed Sensing Selection.....	46
Figure 52: ST MC Workbench – Speed Sensing Selection Modes	46
Figure 53: ST MC Workbench – Current Sensing – FOC 3Sh External amplification	47
Figure 54: ST MC Workbench – Current Sensing – FOC 3 Shunt Internal Amp – Internal Gain	48
Figure 55: ST MC Workbench – Current Sensing – FOC 3 Shunt Internal Amp – External Gain	49
Figure 56: ST MC Workbench – Current Sensing – Two Insulated Current Sensor	50
Figure 57: ST MC Workbench – Current Sensing – FOC 3 Shunt External Amp – HSO.....	51
Figure 58: ST MC Workbench – Current Sensing – FOC 3 Shunt Internal Amp – HSO	52
Figure 59: ST MC Workbench – 6 Step – Current Mode.....	53
Figure 60: ST MC Workbench – 6 Step – Current Mode disabled	53
Figure 61: ST MC Workbench – 6 Step – Current Reading	54
Figure 62: ST MC Workbench – 6 Step – Current Reading not available.....	55
Figure 63: ST MC Workbench – Current Sensing – Over Current Protection External Comparators.....	56
Figure 64: ST MC Workbench – Current Sensing – Over Current Protection Internal Comparators	56
Figure 65: ST MC Workbench – Bus Voltage Sensing	57
Figure 66: ST MC Workbench – Temperature Sensing	58
Figure 67: ST MC Workbench – Speed Sensing – Observer PLL	59
Figure 68: ST MC Workbench – Speed Sensing – Observer Cordic	59
Figure 69: ST MC Workbench – Speed Sensing – High Sensitivity Observer (Sensorless).....	60
Figure 70: ST MC Workbench – Speed Sensing – Quadrature Encoder	61
Figure 71: ST MC Workbench – Speed Sensing with 6 step algo – Hall Sensor.....	62
Figure 72: ST MC Workbench – Speed Sensing – Sensorless start-up parameters	63
Figure 73: ST MC Workbench – Speed Sensing – HSO Sensorless start-up parameters	64
Figure 74: ST MC Workbench – 6 Step – Sensorless ADC.....	65
Figure 75: ST MC Workbench – 6 Step – Sensorless – Start-up parameters	66
Figure 76: ST MC Workbench –Driver Protection.....	67
Figure 77: ST MC Workbench – Drive Settings – Speed Control	68
Figure 78: ST MC Workbench – Drive Settings – Torque Control	69
Figure 79: ST MC Workbench – Drive Settings – Position Control	70
Figure 80: ST MC Workbench – Drive Settings – FOC Speed Control (for HSO)	71
Figure 81: ST MC Workbench – Drive Settings – FOC Torque Control (for HSO)	72
Figure 82: ST MC Workbench – Drive Settings – 6 Step	73
Figure 83: ST MC Workbench – Stage Configuration.....	74

Figure 84: ST MC Workbench – User Interface.....	75
Figure 85: ST MC Workbench – Application Configuration	76
Figure 86: ST MC Workbench – Pin Usage	77
Figure 87: ST MC Workbench – Hardware Changes	78

1. General information

The ST MC Workbench software tool provides an easy and friendly way to configure your MC application based on your hardware setup.

After the configuration, the user can generate the project file and use it together with STM32CubeMX software for further application extension.

The ST MC Workbench runs on PC systems using Windows® OS, requiring a USB type A connector.

2. Related documents

Available from www.arm.com

- Cortex®-M0 Technical Reference Manual, available from: <http://infocenter.arm.com>
- Cortex®-M3 Technical Reference Manual, available from: <http://infocenter.arm.com>
- Cortex®-M4 Technical Reference Manual, available from: <http://infocenter.arm.com>

Available from www.st.com or your STMicroelectronics sales office

- STM32C0xx datasheets
- STM32F0xx datasheets
- STM32F3xx datasheets
- STM32F4xx datasheets
- STM32G4xx datasheets
- STM32G0xx datasheets
- STM32F7xx datasheets
- STM32H5xx datasheets
- STM32H7xx datasheets
- STM32L4xx datasheets
- STM32U5xx datasheets

3. The ST Motor Control Workbench

Launch the “ST Motor Control Workbench” software tool by clicking either its icon (Cf. Figure 1: ST MC Workbench – Icon) or by running it directly from the installation folder.

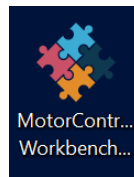


Figure 1: ST MC Workbench – Icon

The first view presented by the ST MC Workbench GUI is a home area (start-up page), containing the following sections:

1. The Toolbar (Id. 1 in Figure 2: ST MC Workbench – GUI (Home View)), used to start a New Project or Load an already existing one, launch the ST Motor Profiler or Motor Pilot software tools, or learn more about software’s documentation. Here It’s also possible to change the language and customize application settings from the Setting button ().
2. The “Recent projects” area (Id. 2 in Figure 2: ST MC Workbench – GUI (Home View)), used to load an already existing project recently modified by the user.
3. The “Examples projects” area (Id. 3 in Figure 2: ST MC Workbench – GUI (Home View)) used to load one of the project samples.

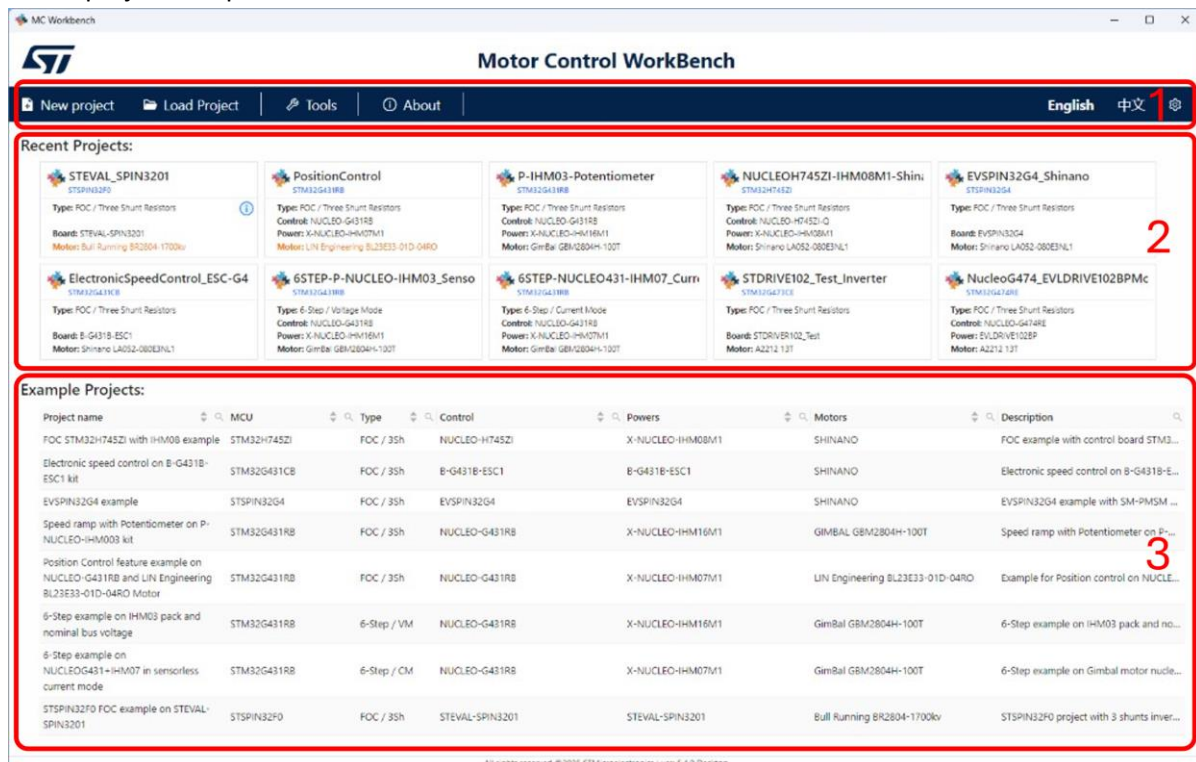


Figure 2: ST MC Workbench – GUI (Home View)

3.1 Editing field behavior

Frequently encountered elements in the *ST MC Workbench* are the *Edit* and *Combo* fields. They present the following format and behavior:

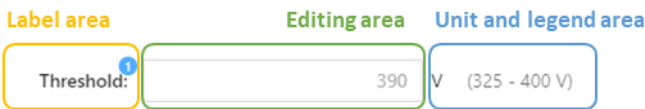
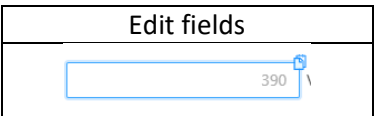


Figure 3: ST MC Workbench – Editing field

- **Label area:** this area contains the field’s name, which may or may not show a numbered notification bullet providing additional info about the field. The color of the bullet specifies its type, that may indicate a *info* (blue), *warning* (orange) or *error* (red) message. The message will pop-up on mouse over.
- **Editing area:** this area lets the user set the field’s value. The field (edit or combo) can assume different styles depending on its editing state:
 - **Field with default value:** no value specified by user (value in light grey). *The default value can be changed silently by the application itself.*

Edit fields	Combo fields
390	ADC1_IN7 (PC1)

- **Field with default value in editing mode** (value in light grey): The user can edit the field’s value. By clicking on the upper-right small icon, the user can transform the value shown in the field from default value to explicit user value (see below).



- **Field with explicit user value** (value in black). *The explicit user value cannot be recalculated by the application.* The user can reset the field to its default value by pressing the upper-right small icon (see below).

Edit fields	Combo fields
391	ADC2_IN7 (PC1)

- **Field with error on value** (value in red): The field needs to be reviewed. The user can edit the value or reset the field to its default value by pressing the upper-right icon.

Edit fields	Combo fields
402	Hall Sensor

- **Unit and legend area:** this area is optional. It indicates the unit and/or the legend of the field (i.e., valid range) (Figure 4: ST MC Workbench – Unit and Legend Area).



Figure 4: ST MC Workbench – Unit and Legend Area

3.2 Hardware Cards

The Hardware Cards (or simply Cards) are frequently used in the whole Workbench. They are UI elements that represent the hardware (Motor, Control, Power, Inverter, Adapter) (Figure 5: ST MC Workbench – Cards (Motor-Control-Power-Inverter-Adapter)). Depending on the context the cards are used in, or the hardware that they represent, the cards can present different actions and information.

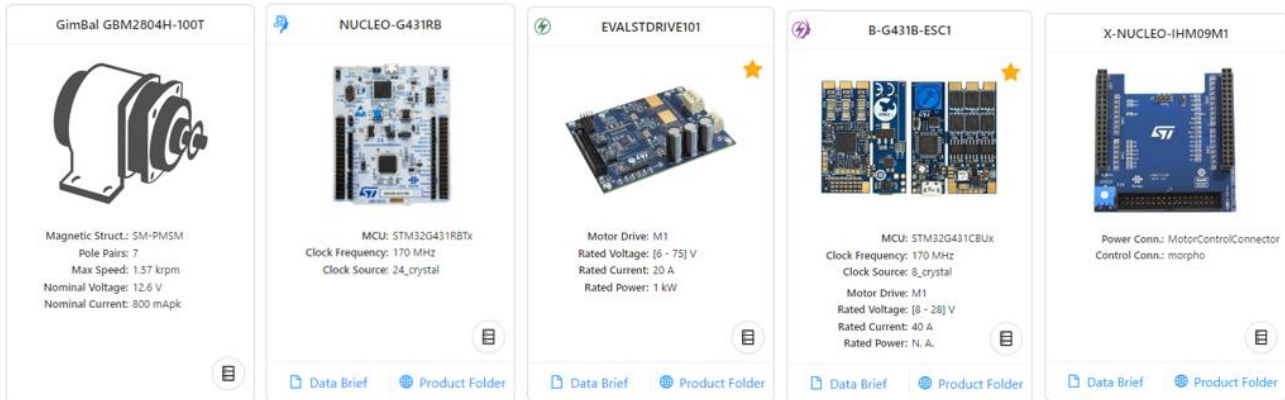


Figure 5: ST MC Workbench – Cards (Motor-Control-Power-Inverter-Adapter)

3.2.1 Cards Information

In most cases the cards consist of four sections:

- the header, which contains the name of the associated hardware and an optional small icon based on hardware's type;
- the image of the physical board or a default one for each hardware type;
- the info area, which contains specific info relative to the hardware;
- the action icon, shown in the middle right;
- the optional footer, which presents two hyperlinks to www.st.com pointing to external documentation about the Data Brief or the Product folder of the board.

3.2.2 Cards Actions

The middle right icon of the card identifies the hardware owner and allows specific actions for the

hardware; the icon  identifies the hardware provided by MC Workbench; the icon  identifies a user custom hardware. The available actions for each kind of hardware:

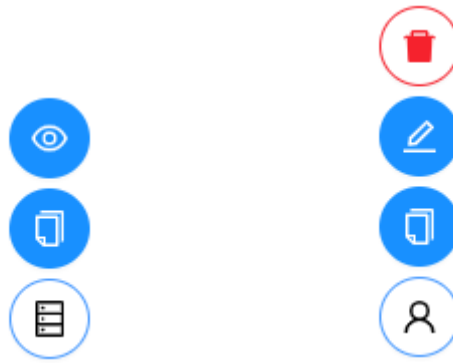


Figure 6: ST MC Workbench – Cards Actions

Hovering on the action icon with the mouse, additional buttons will appear (Figure 6: ST MC Workbench – Cards Actions):

- The *View* button (👁️) lets the user open the related board with the external software *Board Designer* in view mode. This is available with provided hardware only.
- The *Clone* button (📄+) lets the user clone the relative hardware with the external software *Board Designer*. This is available with both provided and user custom hardware.
- The *Edit* button (📄✎) lets the user edit the relative hardware with the external software Board Designer. This is available with user custom hardware only.
- The *Delete* button (🗑️) lets the user delete the relative hardware. This is available with user custom hardware only. **Warning:** the *delete* action cannot be reverted and the Hardware data will be lost; a dedicated warning message will be shown, asking the user to confirm the action (Figure 7: ST MC Workbench – Cards – Delete Custom Hardware Warning).

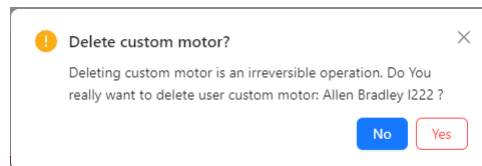


Figure 7: ST MC Workbench – Cards – Delete Custom Hardware Warning

3.2.3 Hardware Preferred

Any kind of hardware can be flagged as preferred using the star icon (★) that appears when the mouse overs the associated card. If the hardware is not flagged as preferred, the star will be empty (☆). Clicking on it will set the relative hardware as preferred, and the star will become always visible, filled with an orange color (★). Clicking it while it is in the preferred state, will remove the hardware from the preferred list, while the aspect and behavior of the star icon will be restored as before. The preferred information changes the order of displayed cards when the relative dialog is refreshed; the cards are sorted first by *preferred* (★), after by then *user* (👤) and last by *provided* (📄) hardware.

3.3 Project Composer Page

The “New Project” button (Cf. Figure 2: ST MC Workbench – GUI (Home View)) leads to the *Composer* page (Figure 8: ST MC Workbench – Composer window). Here the user can define a new project by choosing the hardware and the relative connections step by step. The *Composer* page is a full featured section that helps and guides the user build a new project in an interactive and easy way.

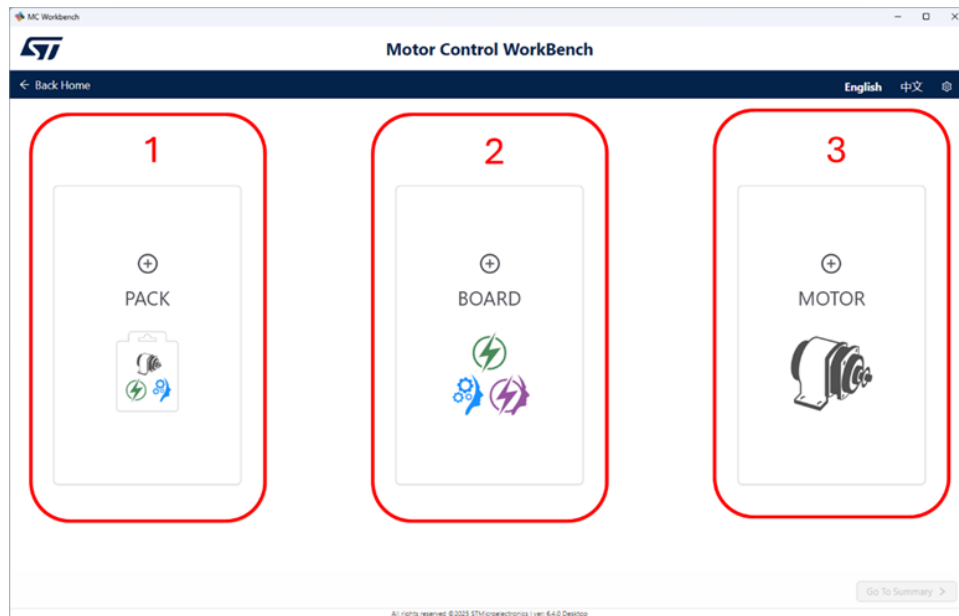


Figure 8: ST MC Workbench – Composer window

3.3.1 Composer Start View

At start, the **Composer Page** lets the user choose whether to start creating the project from a Pack (Id. 1 in Figure 8: ST MC Workbench – Composer window), a Board (Id. 2 in Figure 8: ST MC Workbench – Composer window) or a Motor (Id. 3 in Figure 8: ST MC Workbench – Composer window); there is no preferable order, the user can start from any of the options with no limitation.

3.3.2 Motor

When clicking on **Motor**, a dialog containing the list of available motors pops up (*Figure 9: ST MC Workbench – New Project window – Motors*), allowing the user to select the desired one. In the dialog it is possible to search for a specific motor through the *search field* placed in the right top (id 1 in *Figure 9: ST MC Workbench – New Project window – Motors*). A counter placed next to the field shows the number of visible motors over the available motors. It's also possible to refresh the motor list by pressing the *Refresh* button (⌂) and to import a new motor from the *Import* button (📁).

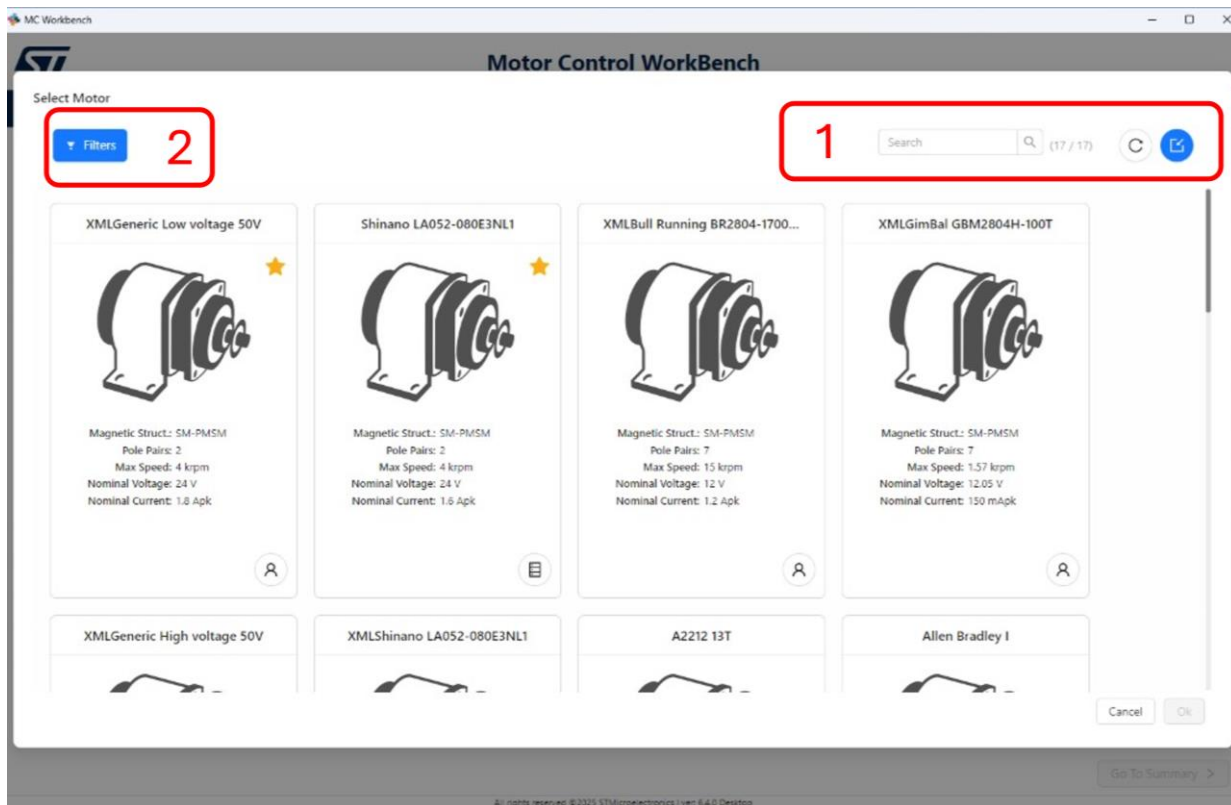


Figure 9: ST MC Workbench – New Project window – Motors

Placed on the top left (Id. 2 in Figure 9: ST MC Workbench – New Project window – Motors), the *Filters* button gives options to filter motors by specific parameters (advanced filter).

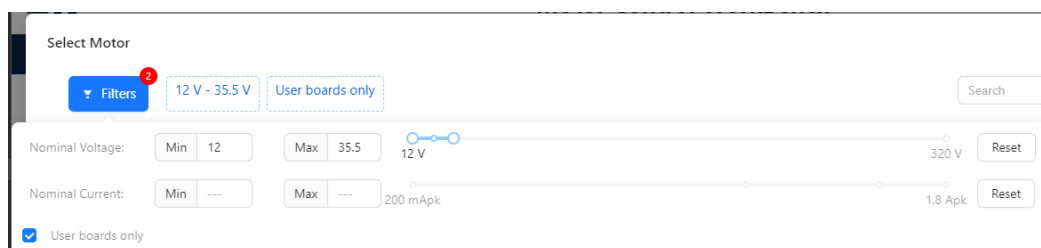


Figure 10: ST MC Workbench – New Project – Motors – Filters

A red bubble (2) keeps count of the active filters; the relative summary is also displayed right next to it on the right side (12 V - 35.5 V User boards only). Clicking the same red bubble resets all active filters, as described by the bubble's UI when hovered by the mouse.

Selecting a motor card with a single click and then pressing the *Ok* button (Ok) placed on bottom right, or simply double clicking a motor card, will finalize the motor's selection, closing the Select Motor dialog and adding the selected motor's card to the main View. The *Cancel* button (Cancel) in the bottom right or the

ESC keyboard's key will close the dialog without applying any changes.

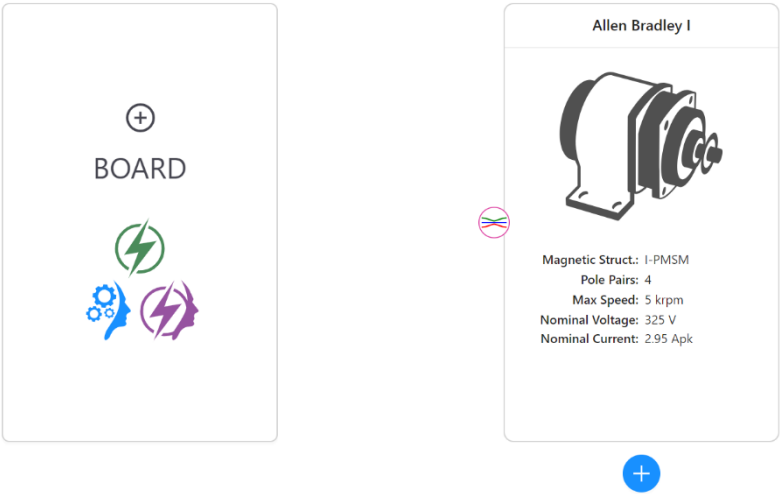


Figure 11: ST MC Workbench – New Project Window – Selected Motor

3.3.2.1 Dual Motor

When a first motor is already been added to the main view, an Add Motor button () will appear under it (Figure 11: ST MC Workbench – New Project Window – Selected Motor). Clicking will start the procedure for the selection of the second motor, similarly to the one described for the first motor. Once the second motor is selected, the *Composer Main View* will appear in as described below (Figure 12: ST MC Workbench – New Project window – Dual Motor selection)

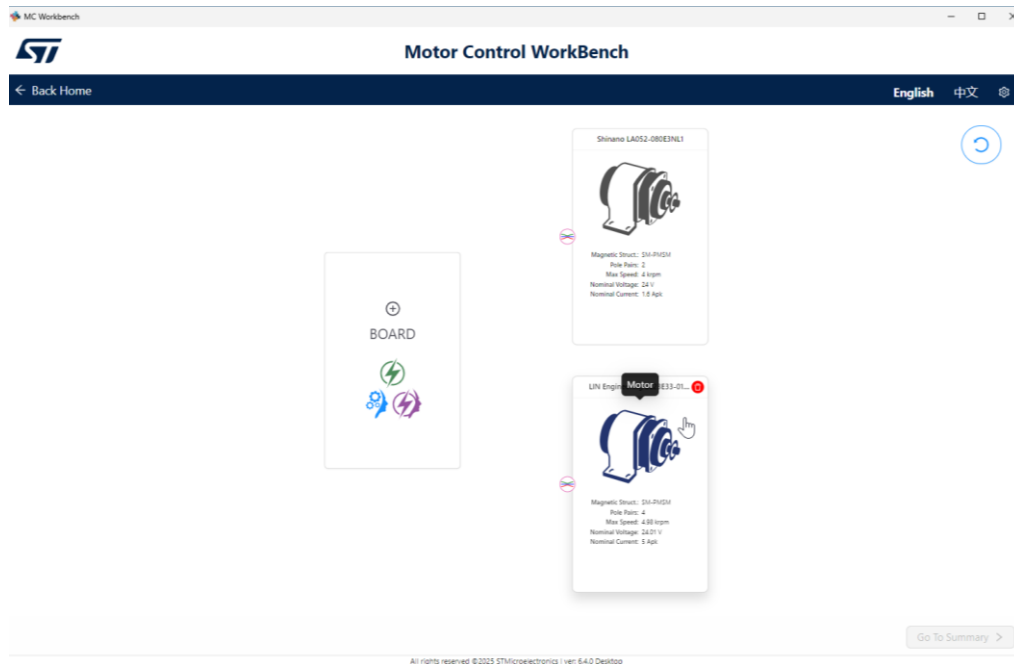


Figure 12: ST MC Workbench – New Project window – Dual Motor selection

It is possible to remove a motor from the main view by clicking the red recycle icon () in the motor card, shown when the mouse overs the card.

3.3.2.2 Clone Motor

All the motors in the list (provided and user custom motors) can be cloned by pressing the Clone icon  , visible in the middle right of motor card when the mouse overs the action icon.

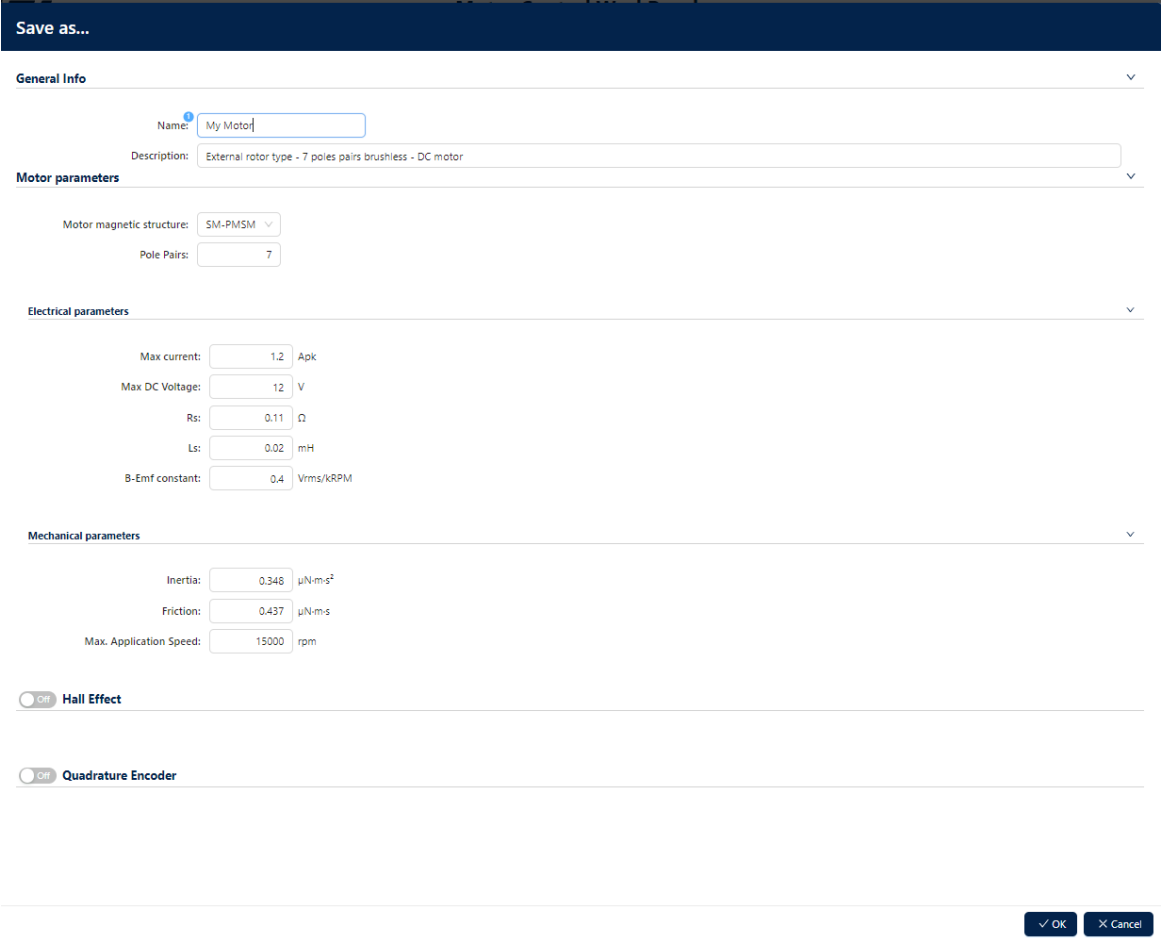


Figure 13: ST MC Workbench – New Project window – Clone/Save as Motor

Clicking the *Clone* button will show a new form (Figure 13: ST MC Workbench – New Project window – Clone/Save as Motor), letting the user customize the new clone motor's parameters. The clone motor will be saved with the name specified in the General Info Name field - assuming that no errors are found – when the **OK** button is pressed. The new clone motor will be shown in the list as a user custom motor. Pressing **Cancel** button will abort the cloning procedure, closing the form.

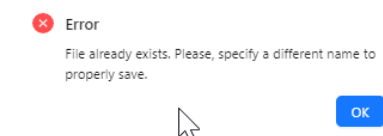


Figure 14: ST MC Workbench – New Project window – Save as Motor Error

When the user clones a custom motor, it is mandatory to specify a different name from the cloned one to avoid data overwrite. A message will pop up otherwise (Figure 14: ST MC Workbench – New Project window – Save as Motor Error)

3.3.2.3 Import Motor

On the top right, next to the search field, the Import button () lets the user import a motor from a file. The click will open a system dialog box allowing the selection of the motor file, which will be validated before being locally stored in a dedicated subfolder inside the user's system user area. If successfully validated, the motor will be correctly added and visible in the list as a "user custom motor".

3.3.3 Board

Starting from **Board** by clicking on it, a *Select Board* dialog will pop up (Figure 15: ST MC Workbench – New Project Window – Board Selection). This dialog will show *Power*, *Control* and *Inverter* cards, all together sorted by type. The user can choose to select one of them in order to add it in the main view.

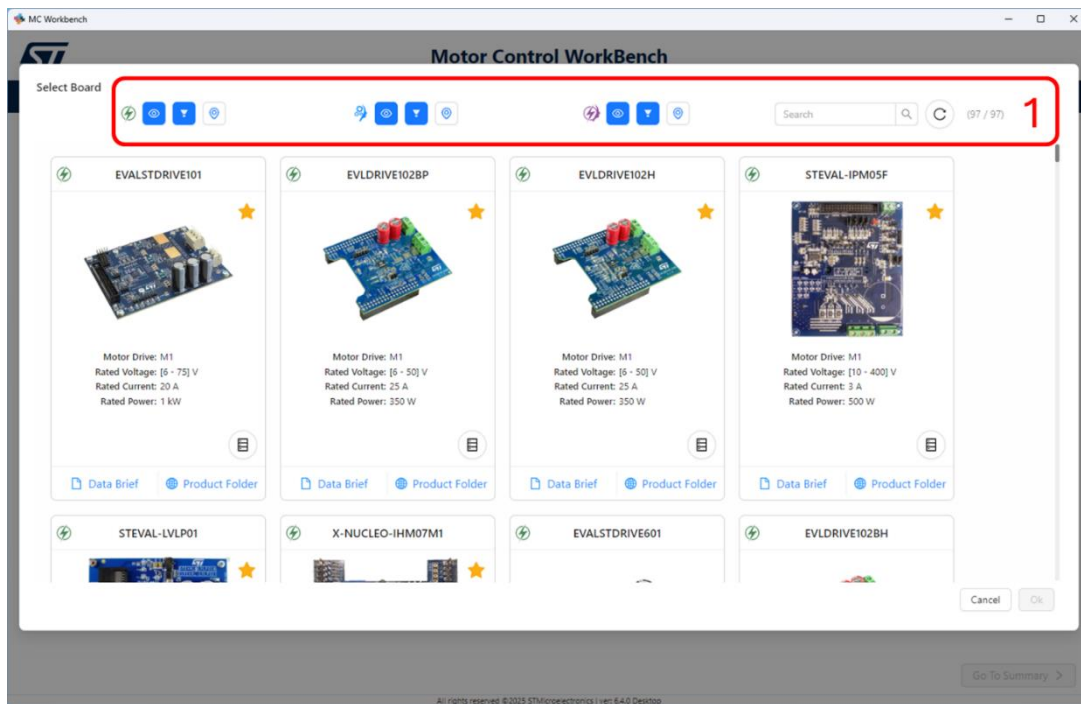


Figure 15: ST MC Workbench – New Project Window – Board Selection

The header of the dialog, placed at the top, consists of:

- the filtering options for each type,
- a single search text field for all the boards,
- the label indicating the currently visible boards over the total number,
- the *Refresh* button () used to refresh the hardware list.

The filtering options are similar for each of the three types. They consist of

- the *Visibility* Button  activate/deactivate the boards' visibility,
- the *Filters* Button  opens a set of advanced filters,
- the *Pin* Button  sets the relative type as the only visible one.

The hardware is displayed on the list using cards, according to the relative type's filters.

3.3.3.1 Power Filters

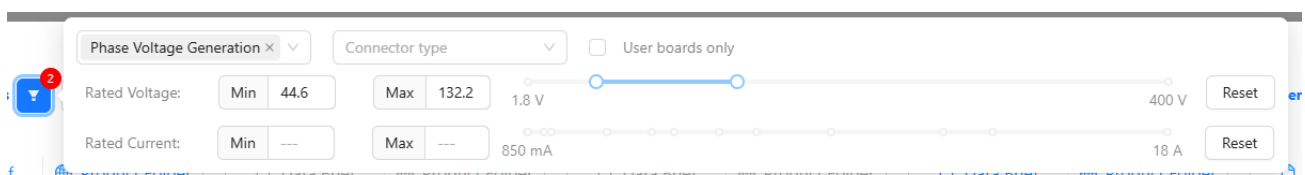


Figure 16: ST MC Workbench – New Project Window – Board Selection Power Filters

The Power Cards displayed in the dialog can be filtered with specific filters for this type of hardware, using the relative filter button placed on the header of the dialog.

Power filters can filter powers based on:

- **Features:** it refers to the power kind of feature that the board can implement. It supports multiple selections in case multiple features are needed in a single board;
- **Connector type:** it refers to the type of connector that the board is provided with. The supported types of connectors are *Morpho* connector, *MotorControlConnector* connector or *PCIe* connector;
- **Rated Voltage:** it refers to the voltage range at which the board is projected to work with;
- **Rated Current:** it refers to the current range at which the board is projected to work with;
- **User boards only:** it filters out the *provided* hardware, showing only the custom *user* boards.

The number of active filters is shown as a red bubble on top of the *Filters* icon. By clicking on it, all the filters can be reset, as shown by the bubble UI when hovered by the mouse. Each filter can also be reset singularly by the relative buttons.

3.3.3.2 Control Filters

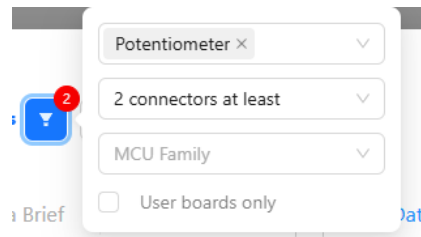


Figure 17: ST MC Workbench – New Project Window – Board Selection Control Filters

The Control Cards displayed in the dialog can be filtered with specific filters for this type of hardware, using the relative filter button placed on the header of the dialog.

Control filters can filter controls based on:

- Features: it refers to the control kind of feature that the board can implement. It supports multiple selections in case multiple features are needed in a single board;
- Number of connectors: it refers to the number of connectors available to connect the board to the powers or adapters. It can have up to two connectors.
- MCU family: it refers to the family of the MCU built in the relative hardware. It supports multiple selection in case different families of control hardware need to be shown at the same time;
- User boards only: it filters out the *provided* hardware, showing only the custom *user* boards.

The number of active filters is shown as a red bubble on top of the *Filters* icon. By clicking on it, all the filters can be reset, as shown by the bubble UI when hovered by the mouse. Each filter can also be reset singularly by the relative buttons.

3.3.3.3 Inverter Filters

The Inverter Cards displayed in the dialog can be filtered with specific filters for this type of hardware, using the relative filter button placed on the header of the dialog.

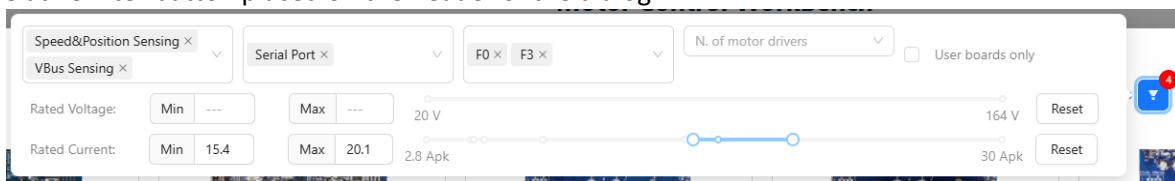


Figure 18: ST MC Workbench – New Project Window – Board Selection Inverter Filters

Inverter filters can filter inverters based on:

- Control Features: it refers to the control kind of feature that the board can implement. It supports multiple selections in case multiple features are needed in a single board;
- Power Features: it refers to the power kind of feature that the board can implement. It supports multiple selections in case multiple features are needed in a single board;
- Number of motor drives: it refers to the number of motor drives available to connect the board to the motors. It can have up to two motor drives;
- MCU family: it refers to the family of the MCU built in the relative hardware. It supports multiple

selection in case different families of hardware need to be shown at the same time;

- Rated Voltage: it refers to the voltage range at which the board is projected to work with;
- Rated Current: it refers to the current range at which the board is projected to work with;
- User boards only: it filters out the *provided* hardware, showing only the custom *user* boards.

The number of active filters is shown as a red bubble on top of the *Filters* icon. By clicking on it, all the filters can be reset, as shown by the bubble UI when hovered by the mouse. Each filter can also be reset singularly by the relative buttons.

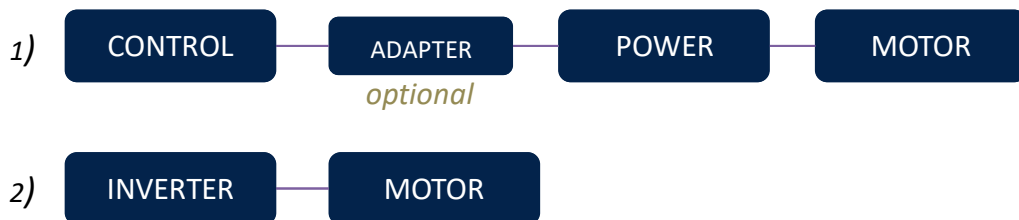
Selecting a card with a single click and then pressing the *Ok* button () placed on bottom right, or simply double clicking a card, will finalize the selection, closing the Select Board dialog and adding the selected hardware to the Main View. The Cancel button () in the bottom right or the ESC keyboard's key will close the dialog without applying any changes.

3.3.4 Composer Flow

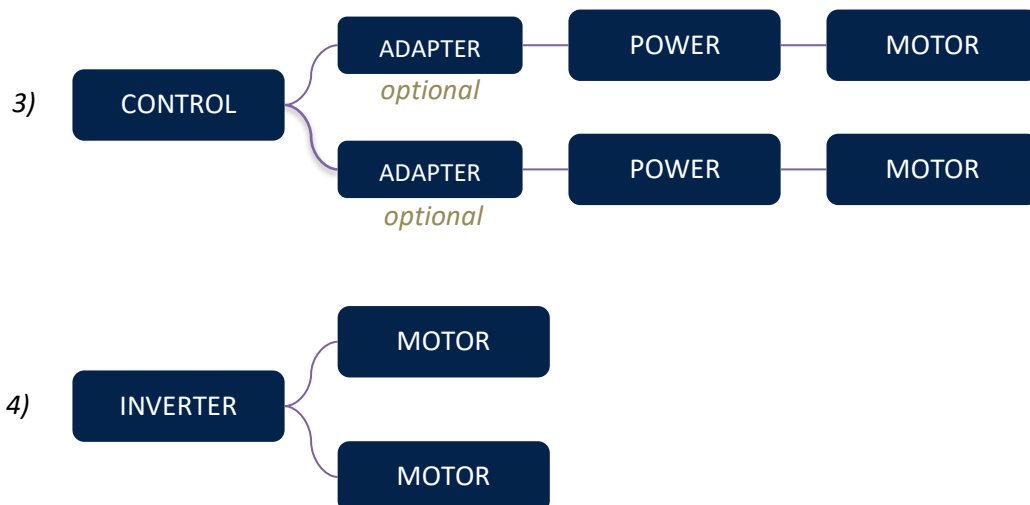
Once the first selection has been made, the process will guide the user through the next steps, towards a correct configuration of the new project.

A correct configuration is a set of hardware composed as one of the following possibilities:

- For Single Motor configurations:



- For Dual Motor configurations:



According to the already selected hardware, the UI will suggest the rest of the hardware needed to reach a correct configuration using placeholders.

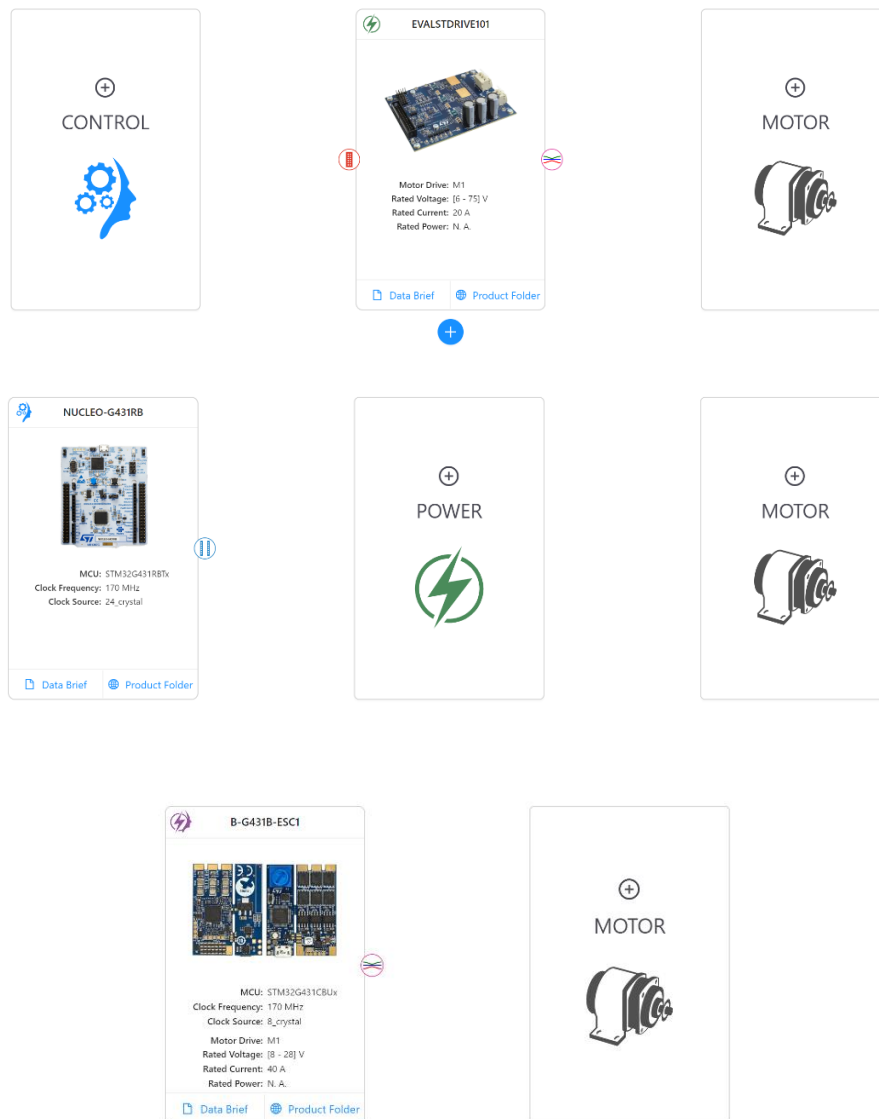


Figure 19: ST MC Workbench – New Project Window – Board Selection Cases

For example, if the first hardware selection is:

- A control board: power and motor placeholders will appear;
- A power board: control and motor placeholders will appear;
- An inverter board: a motor placeholder will appear.

3.3.5.1 Dual Motor

To create a dual motor project, it's sufficient to add a second motor clicking the *Add Motor* button () – as already discussed in previous paragraphs – or add a second power clicking the *Add Power* button, similarly to how it's done for the motor.

3.3.5.2 Adapter Board

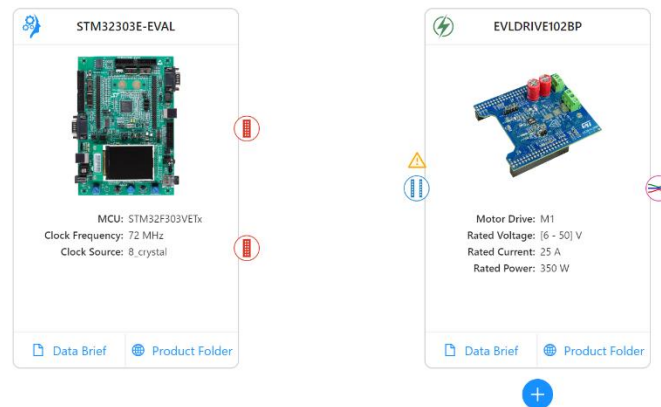


Figure 20: ST MC Workbench – New Project Window – Board Selection Need Adapter

When a control board and a power board are in the scene, their connectors may not be physically compatible. If that's the case, a warning icon () will appear to alert about the necessity of an *Adapter Board* to connect them.

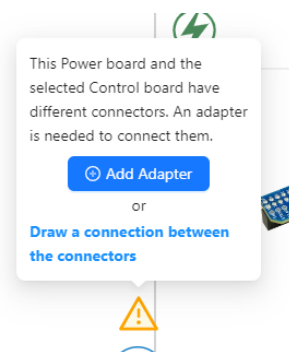


Figure 21: ST MC Workbench – New Project Window – Pop-up for Adapter required

Hovering on the icon with the mouse will show a pop over which describes what's needed and contains instructions to add an Adapter Board. It can be accomplished by pressing the button or by connecting the connectors dragging a line between them. This will open the Adapter Selection Dialog.

3.3.5.2.1 Adapter Selection Dialog

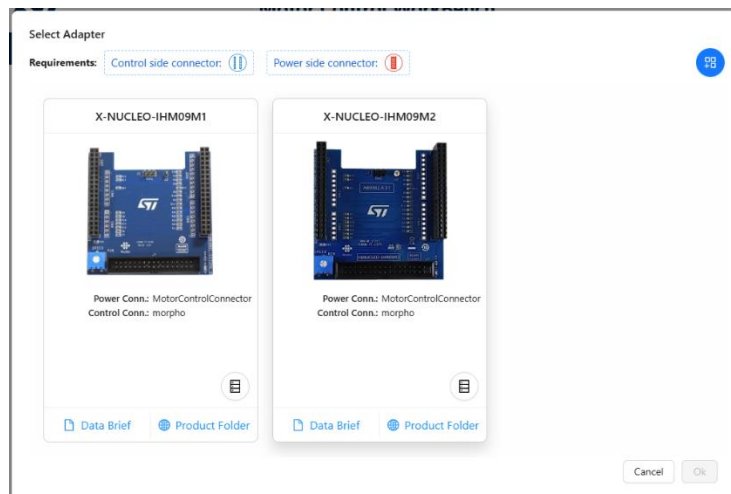


Figure 22: ST MC Workbench – New Project Window – Adapter Selection dialog

The Adapter Selection dialog is divided into two main parts.

- The *header*, which contains info about the connectors that the adapters shown need to have,
- The *Manage Adapters* button that will open the external software “*Board Manager*” to create or modify adapters,
- The *Adapters List* which contains the adapters (provided and custom) whose connectors match the ones needed,

The Adapter Card behaves similarly to the ones already described (3.2).

Clicking on an Adapter card and then pressing the “*Ok*” button on the bottom right – or simply double clicking one of the Adapter Cards – will finalize the selection adding the Adapter Board to the scene.

Clicking on the “*Cancel*” button will close the dialog applying no change.

3.3.5.3 Composer UI features and elements

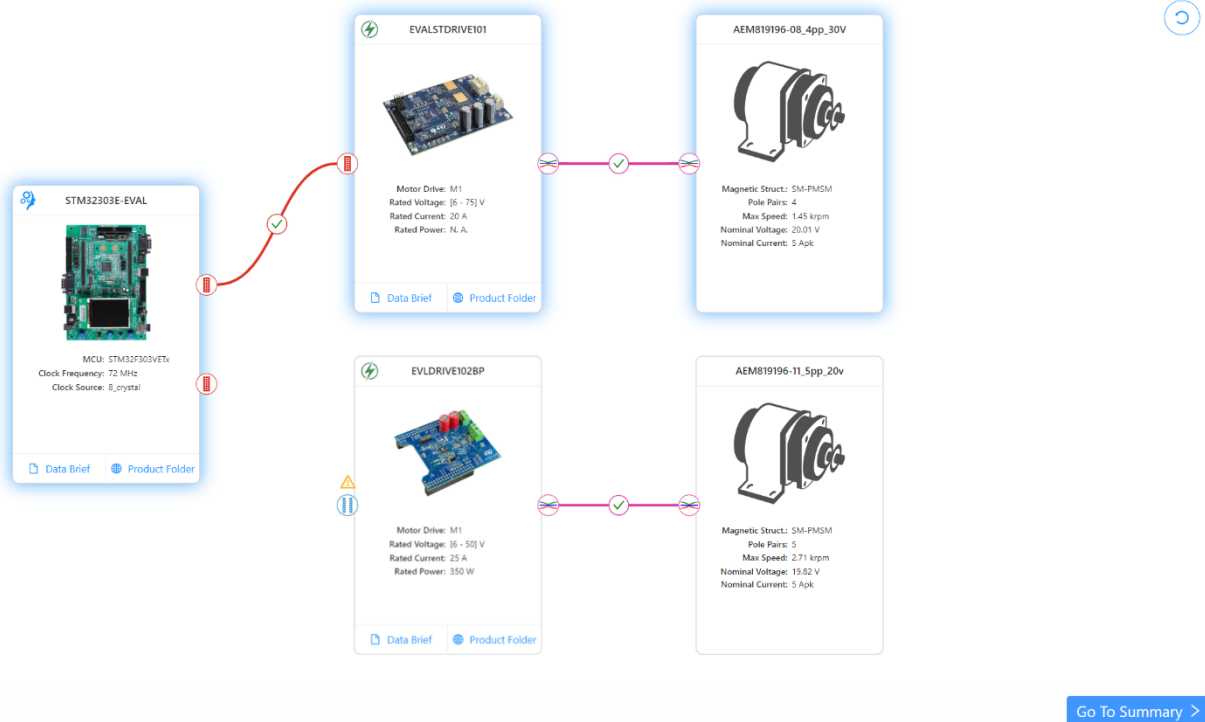


Figure 23: ST MC Workbench – New Project Window – Board Connections

Regardless of what hardware has been added to the scene, when a correct configuration is detected then it will be possible to go to next step (*Summary step*) by clicking on the “Go to Summary” button. Hovering the mouse on the “Go to Summary” button will highlight the hardware in the scene that composes the detected valid configuration.

The *Reset Configuration* button (⌂) will pop up on the upper right side of the scene (if at least one hardware element has been selected) to remove all selections made.

When a hardware element is added or changed, the Composer tool will automatically update its connections using the first compatible connectors found.

3.3.5.4 Connections

Hardware elements in the Composer’s scene need to be connected to compose a proper configuration.

The common way to connect two hardware elements is to drag a connection between two connectors of the same type.

The connectors may be of type:

- Motor Drive (⚡): it may be part of a Motor or a Power board;
- Morpho (🌀): it may be part of a Power, Adapter or Control board;

- Motor Control Connector (): it may be part of a Power, Adapter or Control board;
- PCI Express (): it may be part of a Power, Adapter or Control board.

Connections may also be performed automatically when adding or editing hardware elements – as already stated in previous paragraphs.

Any connection can be removed using the *Disconnect* button () visible when hovering the mouse on the *Connection Established* icon ().

Connections' main color depend on the type of connectors it's linking.

3.3.5 Pack

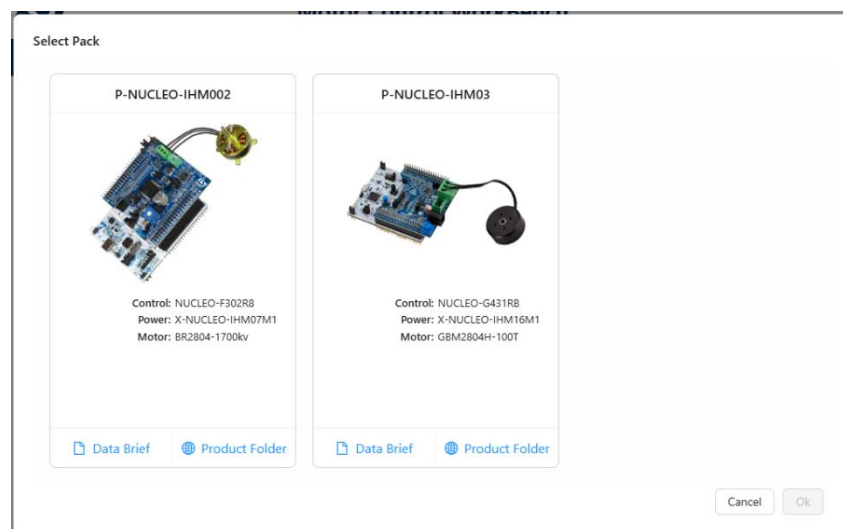


Figure 24: ST MC Workbench – New Project Window – Pack Selection Dialog

Starting the project composition from the *Pack* button, the Pack Selection dialog will open.

A *Pack* is a predefined and preconnected set of hardware provided by ST MC Workbench.

Double clicking a pack card or selecting it and then pressing the *Ok* button will add the hardware in the scene already connected, based on the selected pack info.

It will be possible to edit hardware or connections or just go to the next step (*Summary* step).

3.4 Summary Page

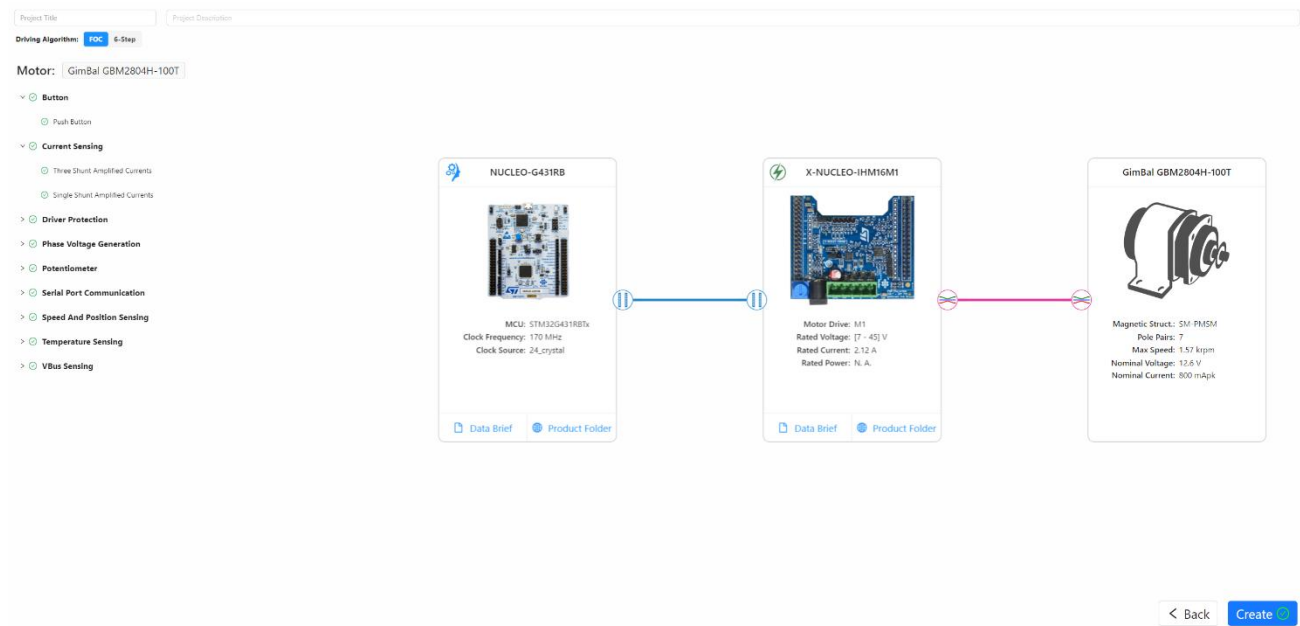


Figure 25: ST MC Workbench – New Project Window – Summary Board Selection and Connections

Clicking on the *Go To Summary* button ([Go To Summary >](#)) will show the *Summary Page* where the full hardware configuration will be analyzed.

The process starts a **deep analysis** of the creating project, going through hardware elements characteristics and how they are connected, understanding what features it can implement and what driving algorithms can be used.

The view is divided into four main parts.

3.4.1 Summary Header

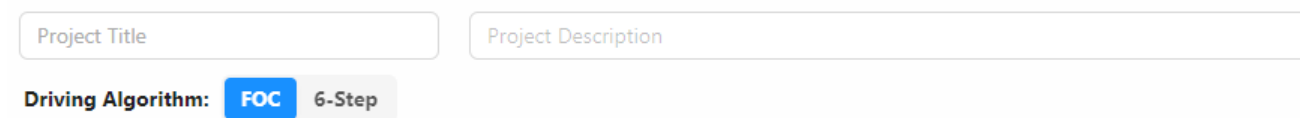


Figure 26: ST MC Workbench – New Project Window – Summary Header

At the top, two optional editable text fields contain the project title and project description. These fields may be left blank.

Right under these two fields, the *Driving Algorithm* section contains a radio button that selects the driving algorithm to use for the creating project. The supported driving algorithms are *FOC* and *6-Step*.



Figure 27: ST MC Workbench – New Project Window – Summary Algorithm not available

Depending on the result of the hardware analysis, any algorithm may or may not be available for the configured project. **If no algorithm is available, it will not be possible to create the project.**

3.4.2 Project's Available Features

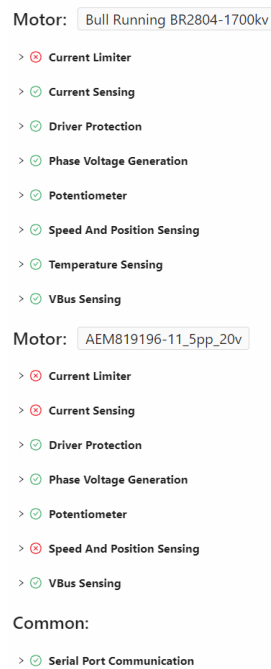


Figure 28: ST MC Workbench – New Project Window – Summary Available features

On the left side of the view, the *features section* shows what features can be implemented using the project.

The features availability is listed for each motor, potentially together with an additional list of features whose availability is not related to any specific motor (Common features).

> Serial Port Communication

Warnings

- IWConnect: BEMFSensingADC BEMF_U is not connected to any matching input (ADC[0-9]*_IN(?IN))
- IWConnect: ExternalCurrentLimiter_PWM PWM_REF is not connected to any matching input (TIM*d+_CH1S)
- CSConnect: ThreeShunt_AmplifiedCurrents: Connection issue for signal CURRENT_AMPL_U->(ADC[0-9]*_IN(?IN))
- SWConnect: SingleShunt_AmplifiedCurrents CURRENT_AMPL is not connected to any matching input ADC[0-9]*_IN(?IN)
- SWConnect: SingleShunt_AmplifiedCurrents_noOffset CURRENT_AMPL is not connected to any matching input ADC[0-9]*_IN(?IN)
- SPConnect: HallEffectSensor No timer matches all signals requirement : HALLSENSOR_H1 HALLSENSOR_H2 HALLSENSOR_H3 -> CH1 CH2 CH3
- SPConnect: QuadratureEncoderRelative No timer matches all signals requirement : QENC_A QENC_B -> CH1 CH2
- PVConnect: QuadratureEncoderAbsolute No timer matches all signals requirement
- IWConnect: BEMFSensingADC BEMF_V is not connected to any matching input (ADC[0-9]*_IN(?IN))
- IWConnect: ExternalCurrentLimiter_PWM PWM_REF is not connected to any matching input (TIM*d+_CH1S)

Figure 29: ST MC Workbench – New Project Window – Summary Warning Report

Under the features list area, technical messages may be displayed depending on the hardware analysis' result. These messages may be simple info messages, warning messages or error messages.

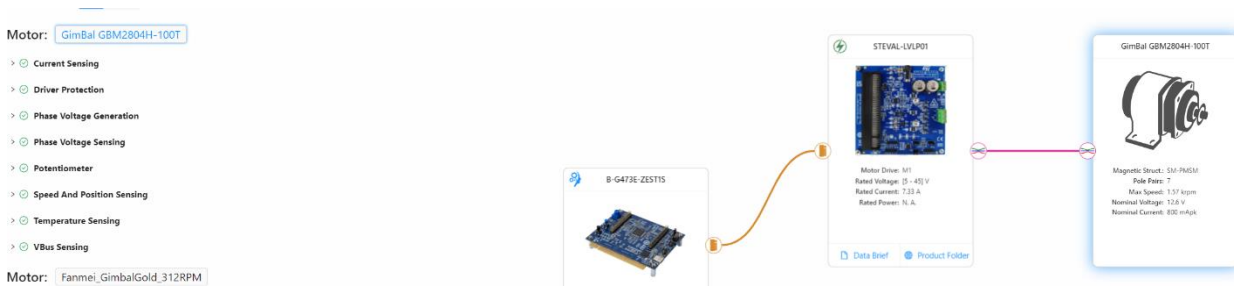


Figure 30: ST MC Workbench – New Project Window – Summary Stage motor

Note: Hovering on a motor name's label will highlight the relative motor component on the *Summary Schema Area*.

3.4.3 Summary Schema

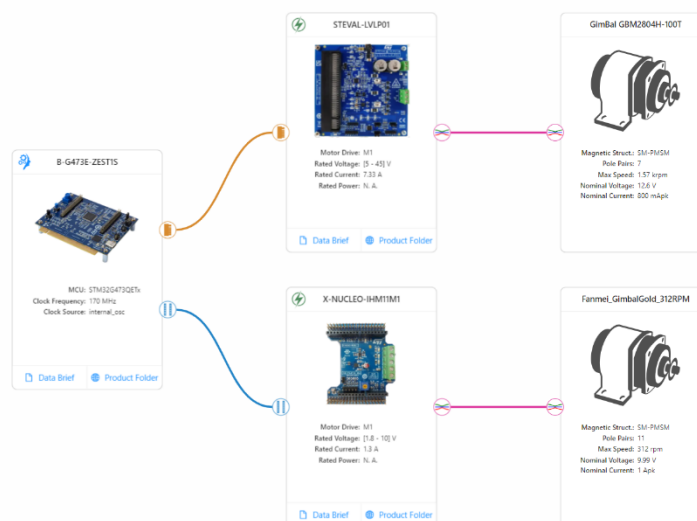


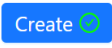
Figure 31: ST MC Workbench – New Project Window – Summary Schema

In the middle-right of the view, the Summary Schema area shows the hardware and connections made in the previews step. It helps the user understand the rest of the information represented in the Summary Page, without allowing any modifications.

3.4.4 Summary Footer

On the bottom of the page, the Summary Footer area contains two buttons positioned on the right side.

The *Back* button () can be used to navigate back to the project composing step.

The *Create* button () can be used to create the project and move to the *Project View*.

3.5 Loading an existing Project

When clicking on the “Load Project” button (Cf. id 1 Figure 2: ST MC Workbench – GUI (Home View)), a system dialog window will be shown letting the user select a project file (.stwb6) to load. Loaded Projects will be shown in the “Project view” (Cf. Figure 32: ST MC Workbench – Project view (global view)) where the user can refine it.

It's also possible to load the IOC project file (CubeMx file (.ioc)) with some limitations:

- The IOC file to load must have been generated by the latest ST MC Workbench (version $\geq 6.2.0$);
- No hardware changes are allowed (enable/disable protections, edit current sensing, ...);
- Only motor(s) and some firmware parameters can be modified;

In any case, in order to have a full control on the project, using a ST MC Workbench (.stwb6) project file is strongly suggested.

3.6 Project view

The project view is composed by the following components:

- A Toolbar area
- A Hardware area info
- A Project Steps area to select which hardware parameters to customize:
 - Motor
 - Power Supply
 - PWM Generation
 - ...
- A Hardware configuration details area: it is used to fine tune the selected hardware functionality and to view all the main info and the protections at a first glance.

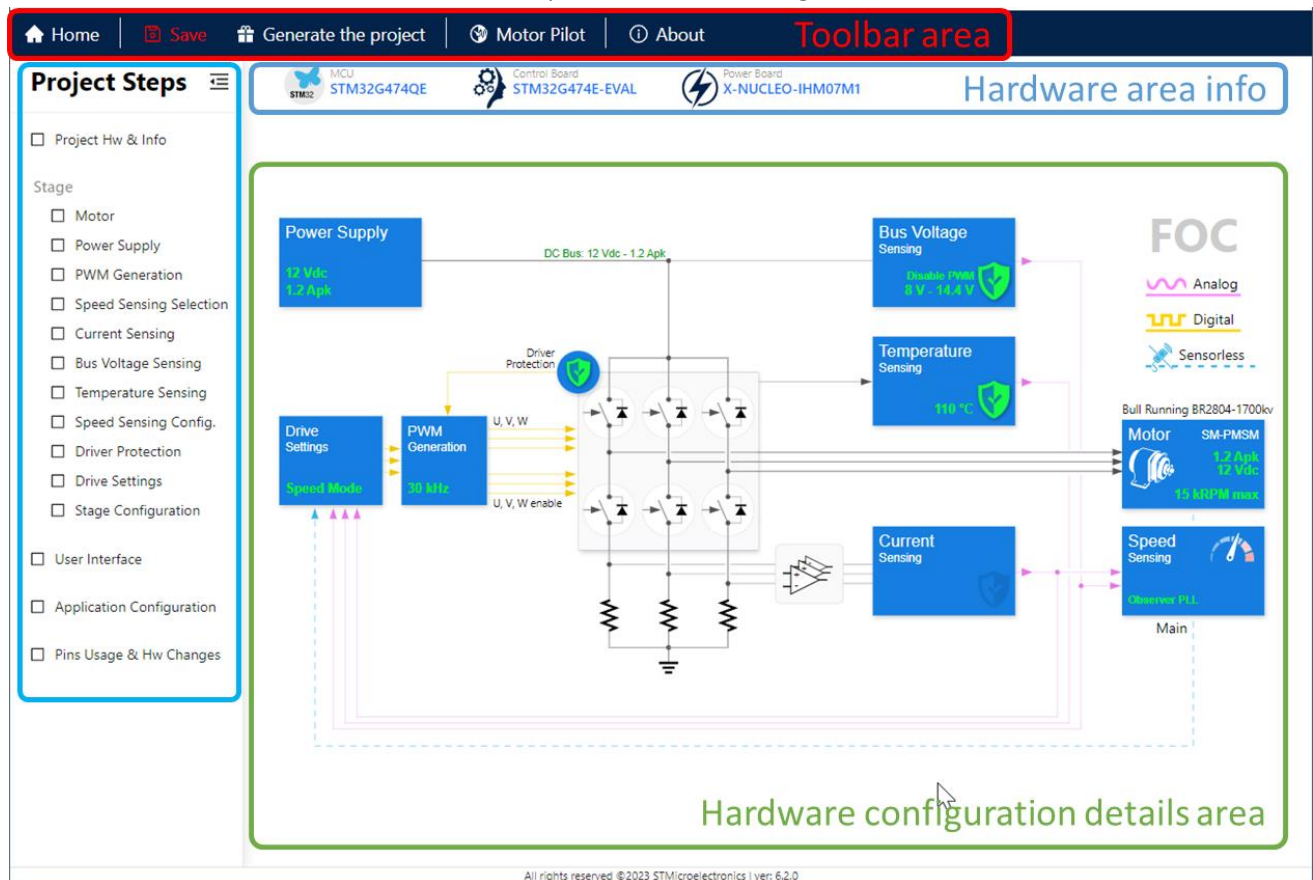


Figure 32: ST MC Workbench – Project view (global view)

3.6.3 Toolbar area

The toolbar area is useful to control application navigation, project saving and generation.

3.6.3.1 Home

The  **Home** button provides the user as easy way to close the current project and go back to the Home View (Cf. Figure 2: ST MC Workbench – GUI (Home View)). In order to do so, if the project contains any changes and has not been saved yet, a confirmation window will pop-up (Cf. Figure 33: ST MC Workbench – Close Project confirmation window) asking the user to save the current project, cancel the action, or going further with the choice of quitting the project without saving it (Cf. Figure 34: ST MC Workbench – Save menu).

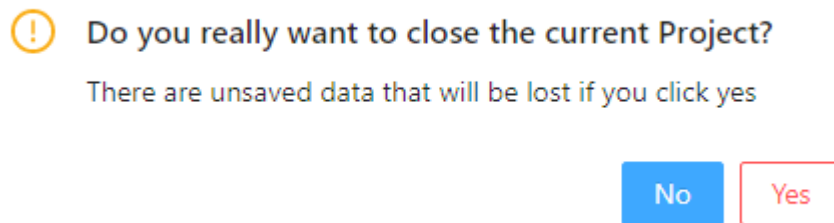


Figure 33: ST MC Workbench – Close Project confirmation window

3.6.3.2 Save

Clicking on  **Save** shows the Save menu.

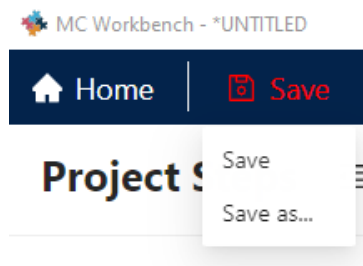


Figure 34: ST MC Workbench – Save menu

- **Save:** save user current project;
If the project has never been saved before, a file manager window pops-up to select the local destination folder for the project file to be saved (Cf. Figure 35).
- **Save as...** save user project as new one file;
A file manager window will always pop-up to save user project as a new one (Cf. Figure 35).

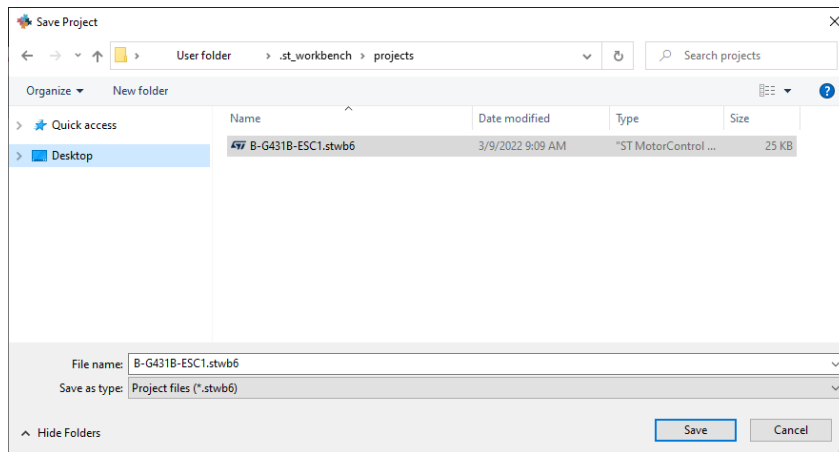


Figure 35: ST MC Workbench – Project Save As window

3.6.3.3 Generate the project

Clicking on the **Generate the project** button will start the process of user application project files generation. If the project file has some has not already been saved, an information window will pop-up to indicate that this project needs to be saved before the generation. Use the “Save & Generate” button to save it and start generating.

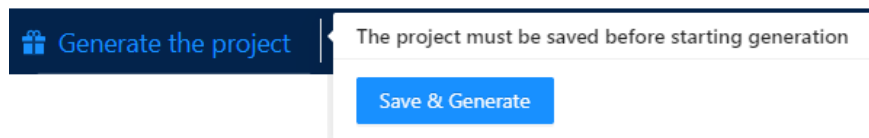


Figure 36: ST MC Workbench – Save & Generate

If some hardware change needs depending on the hardware and the configuration used in the project, an information window pop-up with the required changes (Figure 37: ST MC Workbench – Generate – Hardware Changes). The user can continue with the generation phase or cancel returning to the project window to refine the project setting.

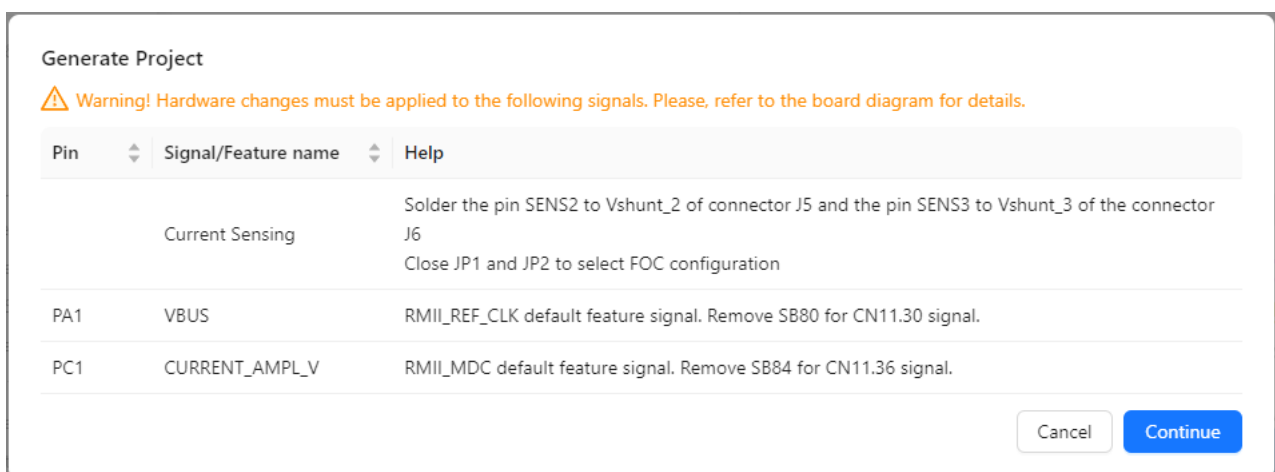


Figure 37: ST MC Workbench – Generate – Hardware Changes

When the generation process starts, a window pops-up (Cf. Figure 38: ST MC Workbench – Project generation) to allow the user select the Target Toolchain, Firmware Package Version, Drive Type and the STM32CubeMx version that will be used to generate the project. Finally, the “GENERATE” button will start the project generation.

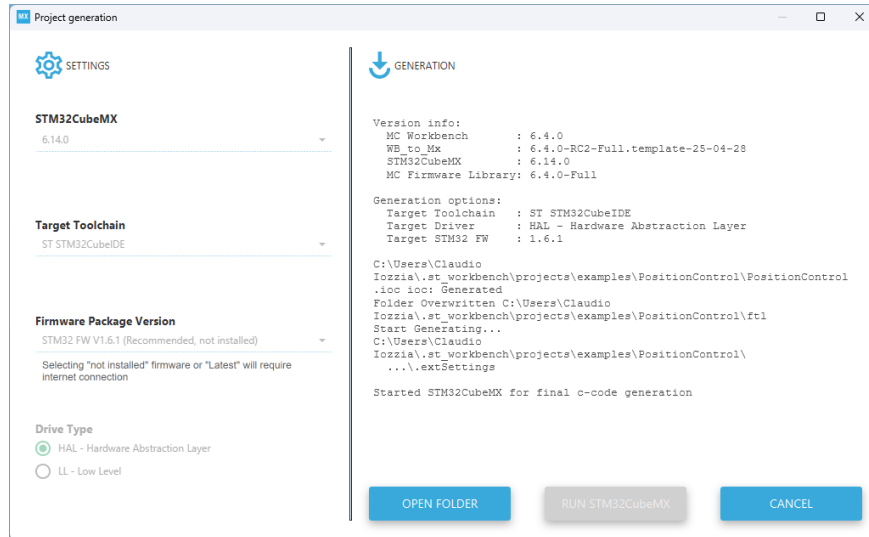


Figure 38: ST MC Workbench – Project generation

3.6.3.4 Motor Pilot

Clicking on Motor Pilot toolbar button, the Motor Pilot application will be launched.

3.6.3.5 About

Clicking on the About menu, a window pops-up showing relevant Workbench information and useful links.

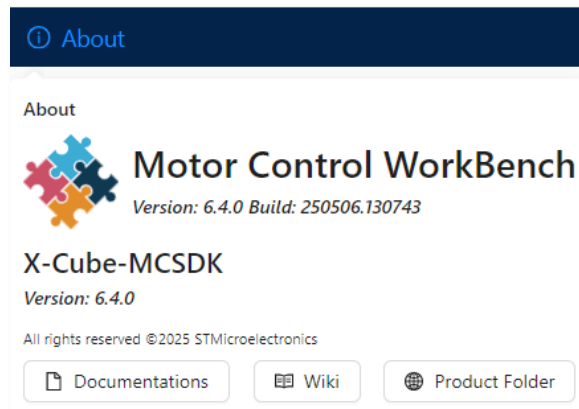


Figure 39: ST MC Workbench – About

3.6.4 Hardware information area

This area will display the main hardware information for the current project.

3.6.4.1 MCU - Pin Usage & Conflicts

Moving over  MCU a window pops-up will be shown (Cf. Figure 40: ST MC Workbench – MCU):

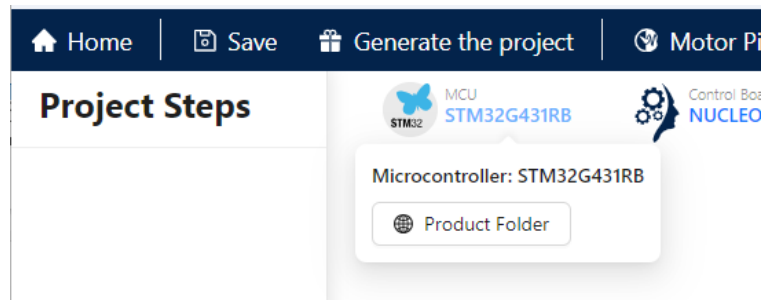


Figure 40: ST MC Workbench – MCU

- **Product Folder**
 - Clicking on it, the Product Folder page for the MCU selected will be open.

3.6.4.2 Control Board

Moving over  Control Board, a window pops-up giving general information about the control board selected. In particular, it shows:

- **Data Brief Folder**
 - Clicking on it, the relative Data Brief will be open.
- **Product Folder**
 - Clicking on it, the relative Product Folder page will be open.

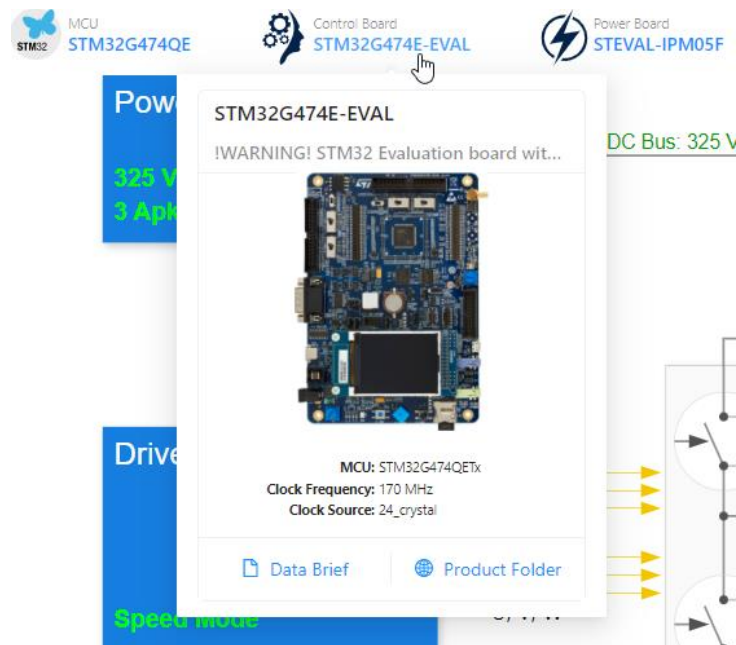


Figure 41: ST MC Workbench – Control Board

3.6.4.3 Power Board

Moving over  Power Board, a window pops-up giving general information about the power board selected. In particular, it shows:

- **Data Brief Folder**
 - Clicking on it, the relative Data Brief page will be open.
- **Product Folder**
 - Clicking on it, the relative Product Folder page will be open.

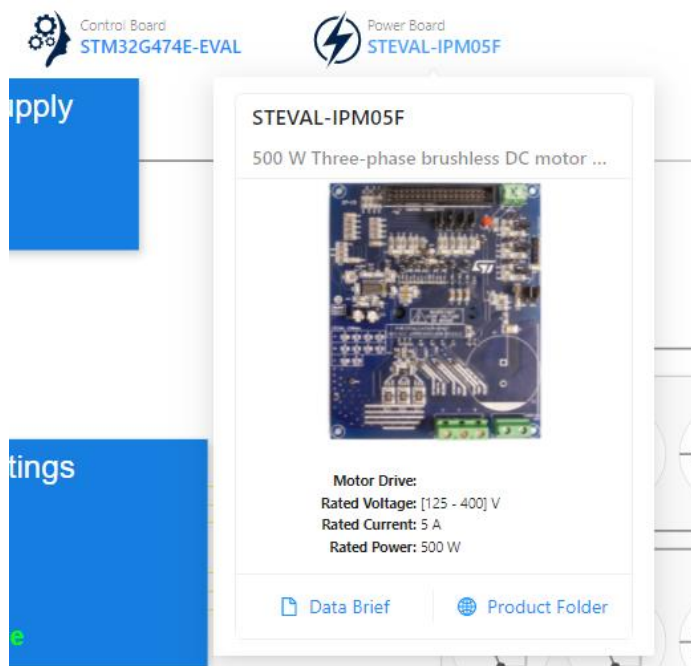


Figure 42: ST MC Workbench – Power Board

3.6.5 Project steps and hardware configuration details

All the hardware configuration relative to the hardware selected are accessible from the wizard steps. The wizard steps are shown in the left application pane (Project Steps area) as well as in the hardware configuration details area (main project view), by clicking on the boxes. The information shown in the displayed schematic reflect the configuration visible in the wizard steps choices.

3.7 Project wizard

The project wizard is accessible from the left pane items or the main schematic view. Clicking on one of these items, the full wizard pop-up and the relative step will be shown. The wizard is customized depending on the selected algorithm, though all the steps are always available having the user being able to freely move over all of them. The project wizard is the smart way to customize and configure the motor control application. Each wizard step contains several checks and info that help the user make the right configurations; in case of errors the wizard doesn't allow to apply the edits made, asking the user to fix the problems before applying them. In any case, the firmware project generation is not allowed if any kind of error is found.

The step sequence on the wizard has a specific meaning: the value of any field in any given step needs to be compatible with all the parameters set in the previous steps (while parameters of the following steps have no importance). The same dependency is also applied within fields inside the same step: the bounding hierarchy always follow the simple “from top to bottom, from left to right” rule.

In case of multi motors application projects, the wizard shows the steps grouped by motor (stage). A “Stage” is a logical concept that identifies a full set composed by the interconnection of the Motor Drive, Power board, eventual connector(s) and/or bridge and Control board, that can make the motor spin.

3.7.3 Project Hw & Info

This is a common step where the user can set the project description, to better identify the project specifying characteristics and extra user information. The project description will be also visible in the recent project section through a small pop-up information icon (Id. 2 in Figure 2: ST MC Workbench – GUI (Home View)).

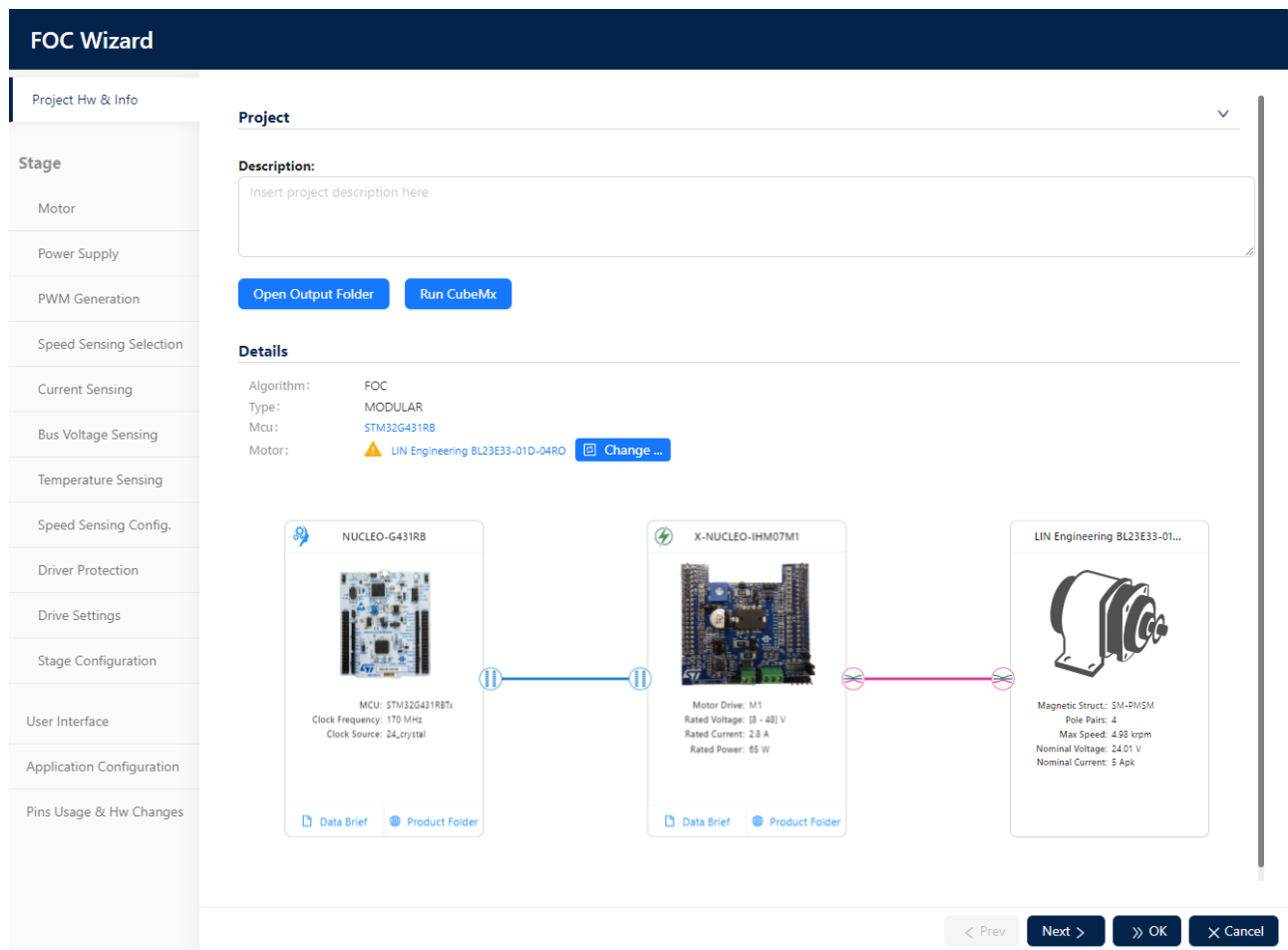


Figure 43: ST MC Workbench – Project Hw & Info

From this step it's possible to open the project output folder and/or launch the CubeMx application if the motor control application project has already been generated before (a project is defined as generated when the relative IOC and firmware source code are properly generated by the ST MC Workbench application). By pressing the **Change ...** button, the user can switch the selected motor(s) choosing from the same list found in the new project creation wizard (both provided or user custom motors) (Figure 44: ST MC Workbench – Project Hw & Info - Change motor). After a switch, it is suggested to review all the steps to check the parameters changes. In the Project HW & Info step there are cards representation of the hardware used and the connection (how hardware is connected). The motor is currently the only hardware switch allowed by ST MC Workbench.

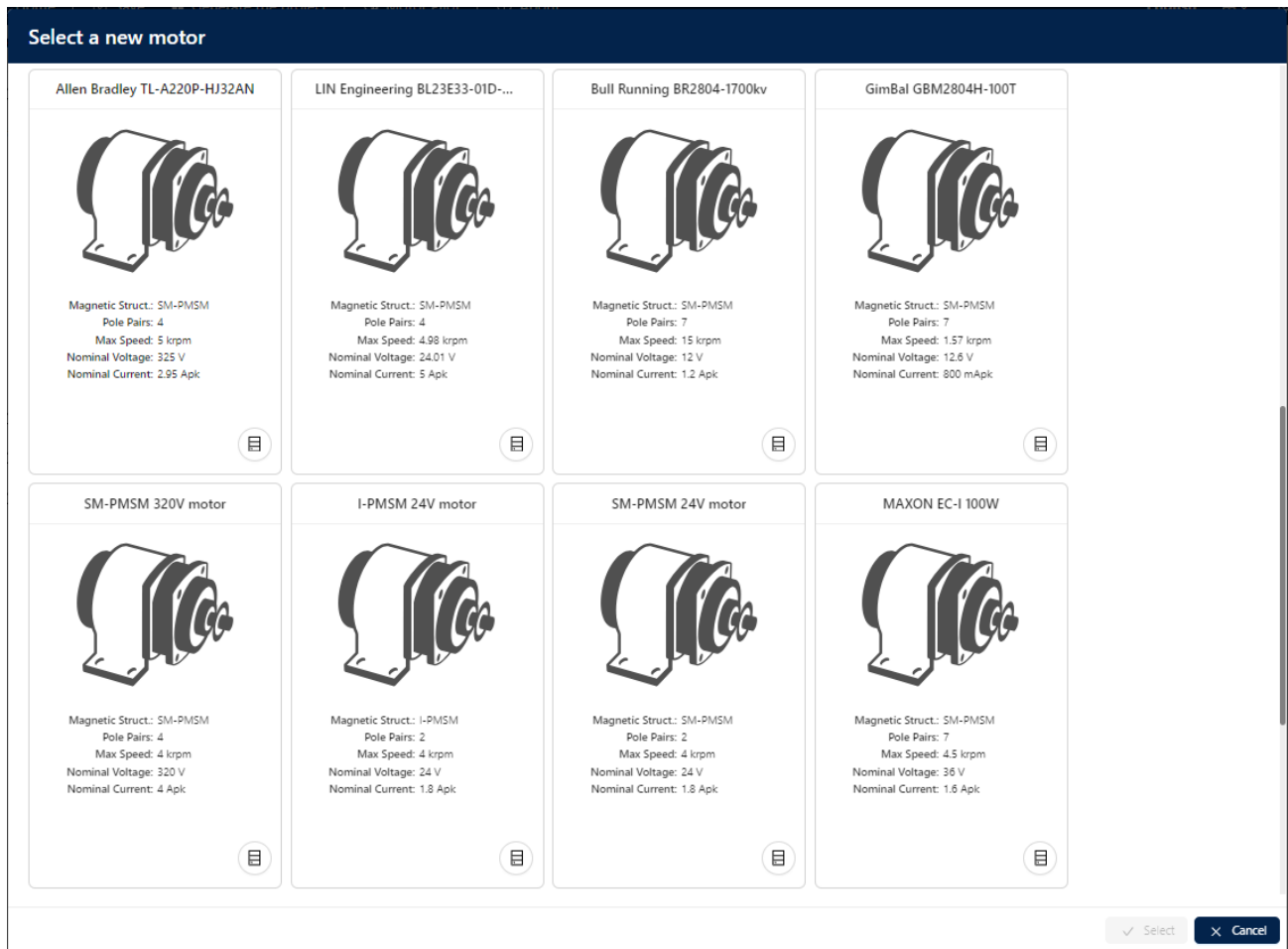


Figure 44: ST MC Workbench – Project Hw & Info - Change motor

3.7.4 Motor

This step is relative to the Motor's parameters, whose default values are loaded during the creation of the project. The user can modify each available parameter according to its value bounds (Cf. Figure 45: ST MC Workbench – Motor), can activate and configure the Hall sensor or Quadrature Encoder for the selected motor (Cf. Figure 46: ST MC Workbench – Motor – Hall Effect – Quadrature Encoder) and change the motor name (this information is used by Motor Pilot application) and save it (throw the save button  Save). The motor saved will be available as a new user custom motor with the defined parameters to be easily used in other motor control application projects.

FOC Wizard

Project Hw & Info

Stage 1

Motor

Power Supply

PWM Generation

Speed Sensing Selection

Current Sensing

Bus Voltage Sensing

Temperature Sensing

Speed Sensing Config.

Driver Protection

Drive Settings

Stage Configuration

Stage 2

Motor

Power Supply

PWM Generation

Speed Sensing Selection

Current Sensing

Bus Voltage Sensing

Temperature Sensing

Speed Sensing Config.

Driver Protection

Drive Settings

Stage Configuration

General Info

Motor name: MAXON EC-I 100W Save

Description: brushless DC motor with encoder and Z Index

Motor parameters

Motor magnetic structure: SM-PMSM

Pole Pairs: 7

Electrical parameters

Max current: 1.6 Apk ✓ Power board maximum rated current: 17 Apk

Max DC Voltage: 36 V ✓ Power board supported voltage range: (10 - 74 Vdc)

Rs: 0.43 Ω

Ls: 0.338 mH

B-Emf constant: 6.14 Vrms/kRPM

Rated Flux: 0.042971 V/Hz

Mechanical parameters

Inertia: 5 $\mu\text{N}\cdot\text{m}\cdot\text{s}^2$

Friction: 14 $\mu\text{N}\cdot\text{m}\cdot\text{s}$

Max. Application Speed: 4500 rpm

On Hall Effect

Sensors displacement: 120

Placement electrical angle: 300

On Quadrature Encoder

Pulses per mechanical revolution: 1024

☒ Has index pin (Ch Z)

< Prev

Next >

>> OK

✕ Cancel

Figure 45: ST MC Workbench – Motor

On Hall Effect

Sensors displacement: 120

Placement electrical angle: 300

On Quadrature Encoder

Pulses per mechanical revolution: 400

☐ Has index pin (Ch Z)

Figure 46: ST MC Workbench – Motor – Hall Effect – Quadrature Encoder

3.7.5 Power Supply

In this step, the Max Current and the Bus Voltage levels that will be used by the application can be set. The values in this step needs to be the one provided from the power supply and must be compatible with the power board and motor.

The screenshot shows the 'FOC Wizard' interface with the 'Power Supply' step selected in the left sidebar. The main area contains the following information:

- Project Info**: Provide here below the Max Current and the Bus Voltage levels that will be used by this application. Please note that these values have to be provided by your Power Supply and they have to be compatible with both PowerBoard and Motor
- Max. application Current**: 2.12 A (Up to 2.12 A)
- Bus Voltage**: 14.8 V (from 7 V to 14.8 V)
- Power board Info**:
 - ✓ Maximum rated current: 2.12 Apk
 - ✓ Supported voltage range: (7 - 45) Vdc
- Motor Info**:
 - ✓ Max current: 5 Apk
 - ✓ Max DC Voltage: 14.8 Vdc

At the bottom right, there are four buttons: '< Prev', 'Next >', '>> OK', and 'X Cancel'.

Figure 47: ST MC Workbench – Power Supply

3.7.6 PWM Generation

In this step, the user can select from the PWM driving topologies available for the selected hardware, the PWM frequency, charge boot capacitor (Time and Duty) and the MCU pin configuration for the topology used. The available configurations depend on the selected driving topology .

3.7.6.1 High and Low Side topology

The following image shows the step for high and low side driving topology.

FOC Wizard

Project Hw & Info

Stage 1

Motor

Power Supply

PWM Generation

Speed Sensing Selection

Current Sensing

Bus Voltage Sensing

Temperature Sensing

Speed Sensing Config.

Driver Protection

Drive Settings

Stage Configuration

Stage 2

Motor

Power Supply

PWM Generation

Speed Sensing Selection

Current Sensing

Config

PWM Frequency: 20000 Hz from 2 kHz to 100 kHz

Driving topology: U, V, W, Uneg, Vneg, Wneg

PWM idle state high side: Turn off

PWM idle state low side: Turn off

SW dead-time: 500 ns from 500 ns to 500 ns

dead-time: 500 ns

Charge Boot Capacitor Time: 10 ms

Charge Boot Capacitor Duty: 0 %

MCU pin mapping

Driver: DRV1

Ch. U: GHS1

Ch. V: GHS2

Ch. W: GHS3

Active Polarity: Active high

Ch. U neg: GLS1

Ch. V neg: GLS2

Ch. W neg: GLS3

Active Polarity: Active high

< Prev

Next >

>> OK

× Cancel

Figure 48: ST MC Workbench - PWM Generation - High and Low Side

3.7.6.2 High Side and One Enable topology

The following image shows the step for high side and one enable driving topology.

FOC Wizard

Temperature Sensing

Speed Sensing Config.

Driver Protection

Drive Settings

Stage Configuration

Stage 2

Motor

Power Supply

PWM Generation

Speed Sensing Selection

Current Sensing

Bus Voltage Sensing

Temperature Sensing

Speed Sensing Config.

Driver Protection

Drive Settings

Stage Configuration

User Interface

Application Configuration

Config

PWM Frequency: 20000 Hz from 2 kHz to 100 kHz

Driving topology: U, V, W, shared enable

PWM idle state high side: Turn off

dead-time: 500 ns

Charge Boot Capacitor Time: 10 ms

Charge Boot Capacitor Duty: 0 %

MCU pin mapping

Timer: TIM8

Ch. U: PA15

Ch. V: PB8-BOOT0

Ch. W: PB9

Active Polarity: Active high

Enable: GPIO_Output (PB7)

Active Polarity: Active high

< Prev

Next >

>> OK

✕ Cancel

Figure 49: ST MC Workbench - PWM Generation – High Side and One Enable

3.7.6.3 High Side and Three Enables topology

The following image shows the step for high side and three enables driving topology.

FOC Wizard

Project Hw & Info

Stage

Motor

Power Supply

PWM Generation

Speed Sensing Selection

Current Sensing

Bus Voltage Sensing

Temperature Sensing

Speed Sensing Config.

Driver Protection

Drive Settings

Stage Configuration

User Interface

Application Configuration

Pins Usage & Hw Changes

Config

PWM Frequency:

30000

Hz

from 2 kHz to 100 kHz

Driving topology:

U, V, W, enU, enV, enW

PWM idle state high side:

Turn off

dead-time:

550 ns

Charge Boot Capacitor Time:

10

ms

Charge Boot Capacitor Duty:

0

%

MCU pin mapping

Timer:

TIM1

Ch. U:

PA8

Ch. V:

PA9

Ch. W:

PA10

Active Polarity:

Active high

Enable Ch. U:

GPIO_Output (PB13)

Enable Ch. V:

GPIO_Output (PB14)

Enable Ch. W:

GPIO_Output (PB15)

Active Polarity:

Active high

< Prev

Next >

>> OK

× Cancel

Figure 50: ST MC Workbench – PWM Generation – High Side and Three Enables

3.7.7 Speed Sensing Selection

This step is available for FOC algorithm projects only. The Speed Sensing Selection is the pre-selection step for the Main Speed Sensing Mode, as in this step it's possible to select the main sensing mode later used in the Speed Sensing Config. Step; this selection has a relevant impact on many following steps.

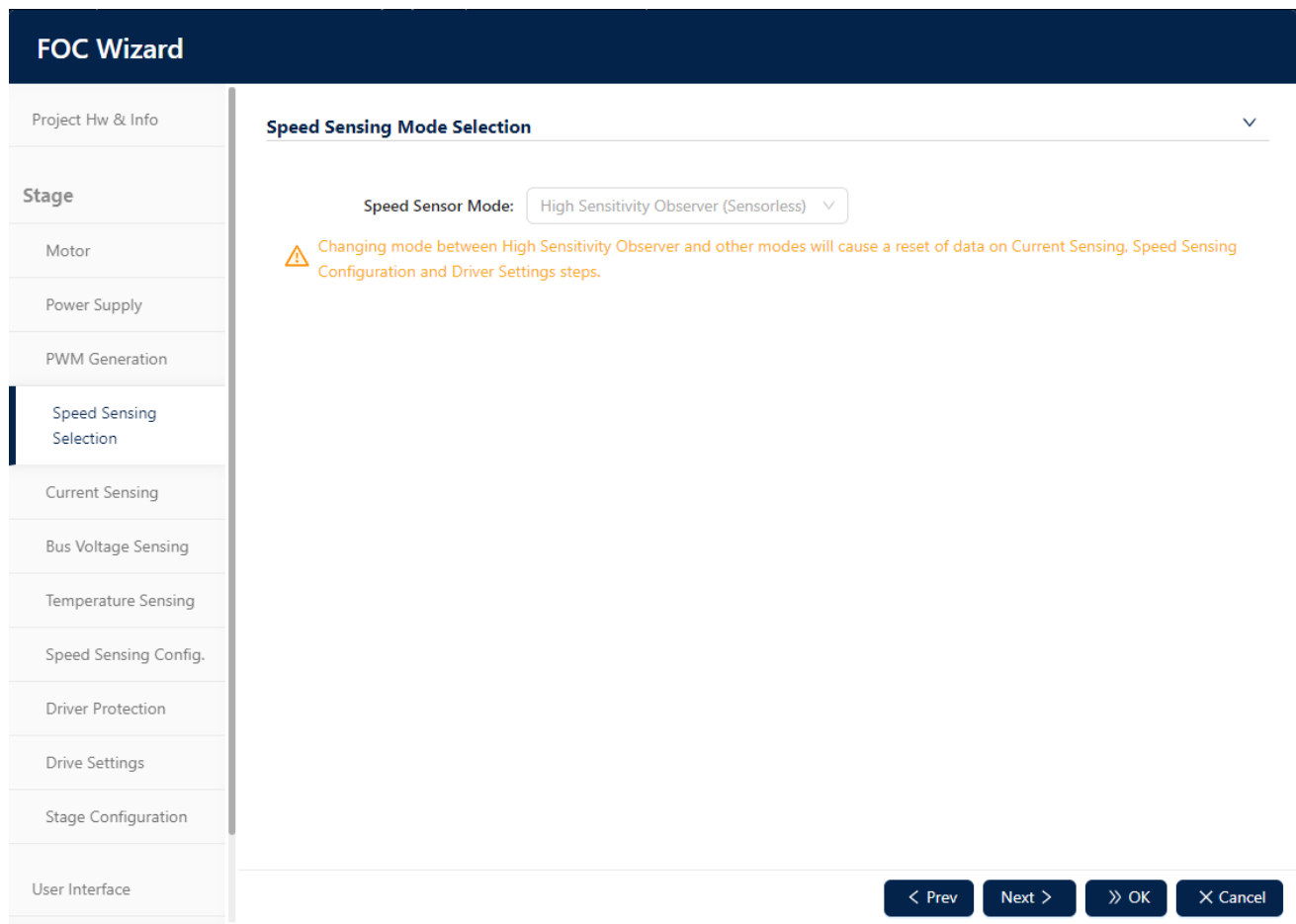


Figure 51: ST MC Workbench – Speed Sensing Selection

All the main speed sensor modes are available from the combo; their availability depend on the hardware used.

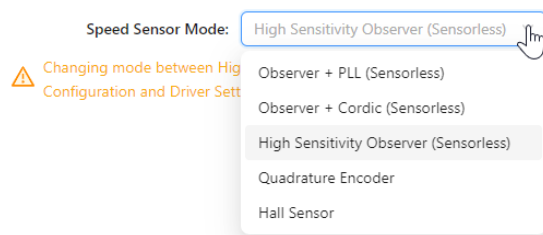


Figure 52: ST MC Workbench – Speed Sensing Selection Modes

The High Sensitivity Observer (a.k.a. HSO) has specific Current sensing, Speed Sensing, Start-up and Drive Settings parameter, so it's available for specific hardware configuration only.

3.7.8 Current Sensing

In this step, the user can choose from the available current reading topologies and amplifications (Cf. Figure 53: ST MC Workbench – Current Sensing). Depending on the topology, amplification and gain, the user will be prompted with the relative fields to set. Also, the MCU Pin for that configuration will be shown along

with a relative graphical schema. The information on the step also depends on the algorithm selected (FOC or 6 Step).

3.7.8.1 FOC 3 Shunt External Amplification

In this Current Sensing topology, the user can set the Regulator execution time, T-Rise and T-Noise. In this topology, the current amplification happens inside the power board. The following image shows this configuration.

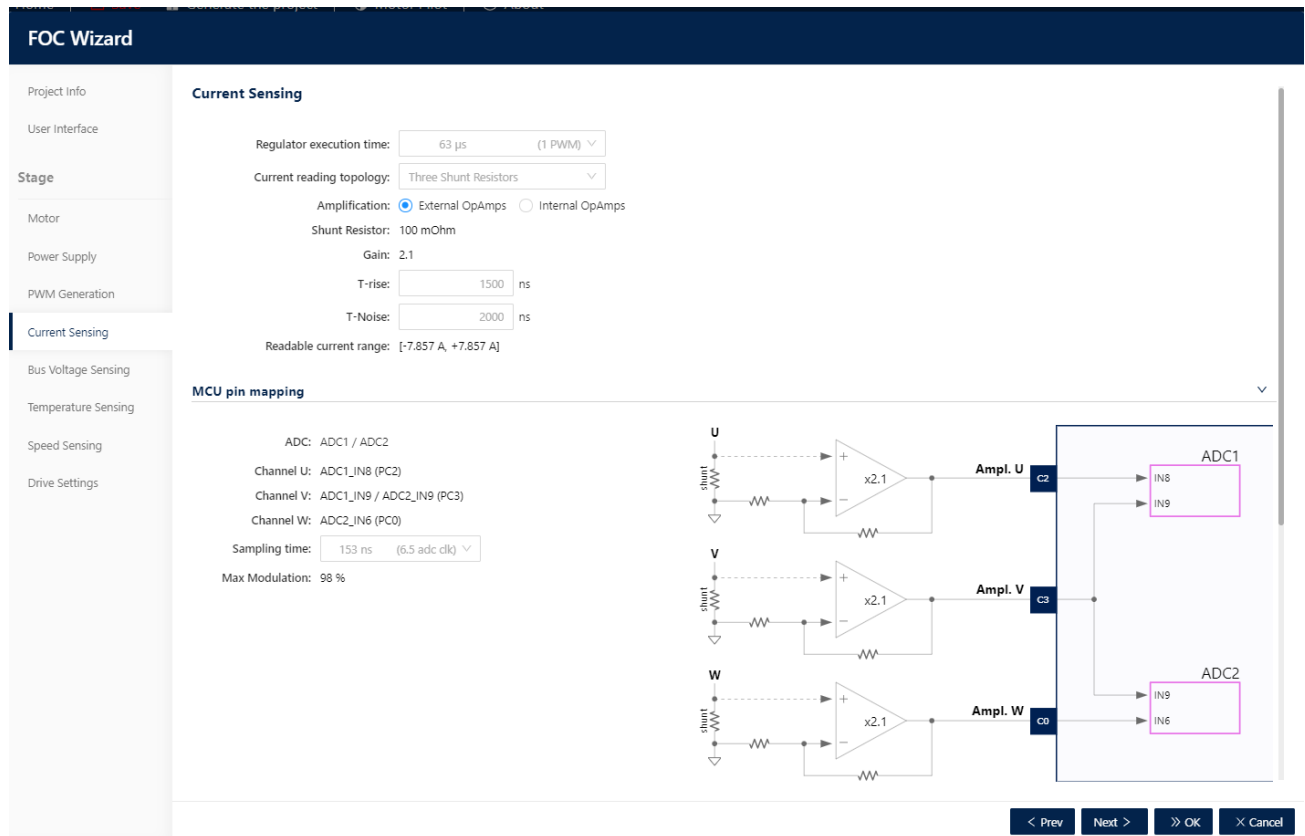


Figure 53: ST MC Workbench – Current Sensing – FOC 3Sh External amplification

3.7.8.2 FOC 3 Shunt Internal Amplification – Internal Gain

In this Current Sensing topology, the user can set the Regulator execution time, internal PGA Gain factor, T-Rise and T-Noise. In this case, the current amplification is executed using internal MCU OpAmps. The following image shows the configuration.

FOC Wizard

Project Info

User Interface

Stage

Motor

Power Supply

PWM Generation

Current Sensing

Bus Voltage Sensing

Temperature Sensing

Speed Sensing

Drive Settings

Current Sensing

Regulator execution time: 63 μ s (1 PWM)

Current reading topology: Three Shunt Resistors

Amplification: ☐ External OpAmps ☒ Internal OpAmps

Gain type: Internal PGA

Sensing Type: Single Ended

Shunt Resistor: 100 mOhm

Internal PGA Gain: 4 1.65 V

T-rise: 2550 ns

T-Noise: 2000 ns

Readable current range: [-4.72 A, 4.71 A] $\rightarrow$ $\pm$ 4.71 A

MCU pin mapping

OpAmp: OPAMP1 / OPAMP2

Channel U: OPAMP1_VINP (PA1)

Channel V: OPAMP1_VINP / OPAMP2_VINP (PA7)

Channel W: OPAMP2_VINP (PD14)

ADC: ADC1 / ADC2

OpAmp1 Out: ADC1_IN3 (PA2)

OpAmp2 Out: ADC2_IN3 (PA6)

Sampling time: 153 ns (6.5 adc clk)

Max Modulation: 96 %

On Over Current Protection

U

shunt

W

shunt

V

shunt

OPAMP1

x3.5

ADC1

IN3

OPAMP2

x3.5

ADC2

IN3

Int. PGA

Int. PGA

< Prev

Next >

>> OK

× Cancel

Figure 54: ST MC Workbench – Current Sensing – FOC 3 Shunt Internal Amp – Internal Gain

3.7.8.3 FOC 3 Shunt Internal Amplification – External Gain

In this Current Sensing topology, the user can set the Regulator execution time, T-Rise and T-Noise. In this case, the current amplification is executed using internal MCU OpAmps, while the gain depends on the gain net inside the Control board. The following image shows this configuration.

FOC Wizard

Project Info

User Interface

Stage

Motor

Power Supply

PWM Generation

Current Sensing

Bus Voltage Sensing

Temperature Sensing

Speed Sensing

Drive Settings

Current Sensing

Regulator execution time: 63 μ s (1 PWM)

Current reading topology: Three Shunt Resistors

Amplification: ☐ External OpAmps ☒ Internal OpAmps

Gain type: External gain net

Sensing Type: Single Ended

Shunt Resistor: 100 mOhm

Gain: 3.5

T-rise: 2550 ns

T-Noise: 2000 ns

Readable current range: [-4.72 A, 4.71 A] $\rightarrow$ $\pm$ 4.71 A

MCU pin mapping

OpAmp: OPAMP1 / OPAMP2

Channel U Inv: OPAMP1_VINM (PA3)

Channel V Inv: OPAMP1_VINM (PA3) / OPAMP2_VINM (PA5)

Channel W Inv: OPAMP2_VINM (PA5)

Channel U: OPAMP1_VINP (PA1)

Channel V: OPAMP1_VINP / OPAMP2_VINP (PA7)

Channel W: OPAMP2_VINP (PD14)

ADC: ADC1 / ADC2

OpAmp1 Out: ADC1_IN3 (PA2)

OpAmp2 Out: ADC2_IN3 (PA6)

Sampling time: 153 ns (6.5 adc clk)

Max Modulation: 96 %

< Prev

Next >

>> OK

× Cancel

Figure 55: ST MC Workbench – Current Sensing – FOC 3 Shunt Internal Amp – External Gain

3.7.8.4 FOC Two Insulated Current Sensor

This Current Sensing mode is available for FOC projects only, if the relative hardware variant is available for the selected hardware.

FOC Wizard

Project Hw & Info

Stage

Motor

Power Supply

PWM Generation

Speed Sensing Selection

Current Sensing

Bus Voltage Sensing

Temperature Sensing

Speed Sensing Config.

Driver Protection

Drive Settings

Stage Configuration

User Interface

Application Configuration

Pins Usage & Hw Changes

Current Sensing

Regulator execution time: 50 μ s (1 PWM)

Current reading topology: Two Insulated Current Sensors

Gain: 0.004 V/A

T-rise: 2550 ns

T-noise: 2550 ns

Readable current range: [-412.5 A, 412.5 A] $\rightarrow$ $\pm$ 412.5 A

MCU pin mapping

ADC: 1 - ADC1 / ADC2

ICS Channel U: ADC1_IN8 (PC2)

ICS Channel V: ADC2_IN9 (PC3)

Sampling time: 153 ns (6.5 adc clk)

U

4 mV/A

Phase U

C2

ADC1

IN8

V

4 mV/A

Phase V

C3

ADC2

IN9

On Over current protection

OCP Mode: External Comparators

Digital Filter Duration: 47.06 ns

Threshold: 200 A

Signal polarity: Active low

Timer: TIM1_BKIN (PE15)

U V W

Shunt

Gain Threshold 200 A

OP Trigger

E15

TIM1

BKIN

< Prev

Next >

>> OK

X Cancel

Figure 56: ST MC Workbench – Current Sensing – Two Insulated Current Sensor

3.7.8.5 FOC 3 Shunt for HSO

For FOC HSO applications, there are specific configurations that can be done. Here blow the HSO external amplification (REF) and the HSO internal amplification.

The HSO is only available for projects using both FOC algorithm and 3 Shunt Current Sensing.

Project Hw & Info

Stage

Motor

Power Supply

PWM Generation

Speed Sensing Selection

Current Sensing

Bus Voltage Sensing

Temperature Sensing

Speed Sensing Config.

Driver Protection

Drive Settings

Stage Configuration

User Interface

Application Configuration

Pins Usage & Hw Changes

Current Sensing

Regulator execution time:

33 μ s

(1 PWM)

Current reading topology:

Three Shunt Resistors

Amplification:

☐ Internal OpAmps
 ☒ External OpAmps

Shunt Resistor:

15 mOhm

Gain:

15

T-rise:

100

ns

(1 - 10 k ns)

T-noise:

500

ns

(1 - 10 k ns)

Readable current range:

[-7.33 A, 7.33 A]

$\rightarrow \pm 7.33$ A

MCU pin mapping

Channel U (Ampl.):

ADC3_IN8 (PD11)

Channel V (Ampl.):

ADC3_IN9 (PD12)

Channel W (Ampl.):

ADC4_IN11 (PD14)

Sampling time:

153 ns

(6.5 adc clk)

Max Modulation:

100 %

< Prev

Next >

>> OK

✕ Cancel

Figure 57: ST MC Workbench – Current Sensing – FOC 3 Shunt External Amp – HSO

FOC Wizard

Project Hw & Info

Stage

Motor

Power Supply

PWM Generation

Speed Sensing Selection

Current Sensing

Bus Voltage Sensing

Temperature Sensing

Speed Sensing Config.

Driver Protection

Drive Settings

Stage Configuration

User Interface

Application Configuration

Pins Usage & Hw Changes

Current Sensing

Regulator execution time: 33 μ s (1 PWM) ∇

Current reading topology: Three Shunt Resistors

Amplification: ☒ Internal OpAmps ☐ External OpAmps

Gain type: External gain net

Gain: 15

Sensing Type: Differential Filtered ADC Input

Shunt Resistor: 15 mOhm

T-rise: 100 ns (1 - 10 k ns)

T-noise: 500 ns (1 - 10 k ns)

Readable current range: [-7.33 A, 7.33 A] $\rightarrow$ $\pm$ 7.33 A

MCU pin mapping

OpAmp: OPAMP3 / OPAMP4 / OPAMP5

Channel U Inv.: OPAMP3_VINM (PB2)

Channel U: OPAMP3_VINP (PB0)

Channel V Inv.: OPAMP4_VINM (PB10)

Channel V: OPAMP4_VINP (PB11)

Channel W Inv.: OPAMP5_VINM (PB15)

Channel W: OPAMP5_VINP (PB14)

Channel U (Ampl.): OPAMP3_VOUT (PB1)

Channel V (Ampl.): OPAMP4_VOUT (PB12)

Channel W (Ampl.): OPAMP5_VOUT (PA8)

Channel U (Ampl.): ADC3_IN8 (PD11) ∇

Channel V (Ampl.): ADC3_IN9 (PD12) ∇

Channel W (Ampl.): ADC4_IN11 (PD14) ∇

Sampling time: 153 ns (6.5 adc clk) ∇

Max Modulation: 100 %

< Prev

Next >

>> OK

× Cancel

Figure 58: ST MC Workbench – Current Sensing – FOC 3 Shunt Internal Amp – HSO

3.7.8.6 6 Step Current Mode

This is available only for 6 Step algorithm projects. The Current Mode is disabled by the default, the user can freely enable it, given that the feature is available for the selected hardware (C.f. Figure 59: ST MC Workbench – 6 Step – Current Mode), (C.f. Figure 60: ST MC Workbench – 6 Step – Current Mode disabled).

6-Step Wizard

Project Info

User Interface

Stage

Motor

Power Supply

PWM Generation

Current Sensing

Bus Voltage Sensing

Temperature Sensing

Speed Sensing

Drive Settings

Current Sensing

On

Current Mode

Current reading topology: Single Shunt Resistor

Amplification: External OpAmps

Shunt Resistor: 330 mOhm

Gain: 3

Readable current range: [0, 5 A]

MCU pin mapping

PWM Frequency: 60000 from 10 kHz to 100 kHz

Current Reference: TIM3_CH1 (PB4)

ETR: TIM1_ETR (PA12)

Signal polarity: Active high

< Prev

Next >

>> OK

X Cancel

Figure 59: ST MC Workbench – 6 Step – Current Mode

6-Step Wizard

Project Info

User Interface

Stage

Motor

Power Supply

Current Sensing

Off

Current Mode

< Prev

Next >

>> OK

X Cancel

Figure 60: ST MC Workbench – 6 Step – Current Mode disabled

3.7.8.7 6 Step Current Reading

Current Reading is optionally available from Current Sensing step, it depends on the hardware availability (Figure 61: ST MC Workbench – 6 Step – Current Reading available and Figure 62: ST MC Workbench – 6 Step – Current Reading not available when the hardware doesn't support the feature). The Current Reading is required to enable the current monitoring flag (☒ Current Monitoring) and to use the Initial Position Detection (Figure 75: ST MC Workbench – 6 Step – Sensorless – Start-up parameters)

The screenshot displays the '6-Step Wizard' interface in the ST MC Workbench. The left sidebar lists various configuration stages, with 'Current Sensing' currently selected. The main panel is titled 'Current Sensing' and contains the following settings:

- Current Mode:** A toggle switch set to 'Off'.
- Current Reading:** A toggle switch set to 'On', accompanied by a yellow information icon.
- Current reading topology:** Set to 'Single Shunt Resistor'.
- Shunt Resistor:** A text input field containing '0.33' with a unit selector set to Ω .
- Gain:** A text input field containing '3'.
- MCU pin mapping:** A section with a dropdown menu for 'Current Monitoring' set to 'ADC1_IN7 (PC1)' and a 'Sampling time' dropdown set to '153 ns (6.5 adc clk)'.
- Circuit Diagram:** A schematic showing a shunt resistor (labeled 'shunt 330 m Ω ') connected to a motor terminal 'U'. The other end of the shunt resistor is connected to ground. The voltage across the shunt is measured by an op-amp configured as a non-inverting amplifier with a gain of 'x3'. The output of the amplifier is labeled 'Ampl. Current' and is connected to a pin labeled 'C1'. This pin is connected to the 'IN7' input of an ADC block labeled 'ADC1'.
- Current Monitoring:** A toggle switch at the bottom of the diagram area, set to 'On'.

At the bottom right of the wizard, there are four navigation buttons: '< Prev', 'Next >', '>> OK', and 'X Cancel'.

Figure 61: ST MC Workbench – 6 Step – Current Reading

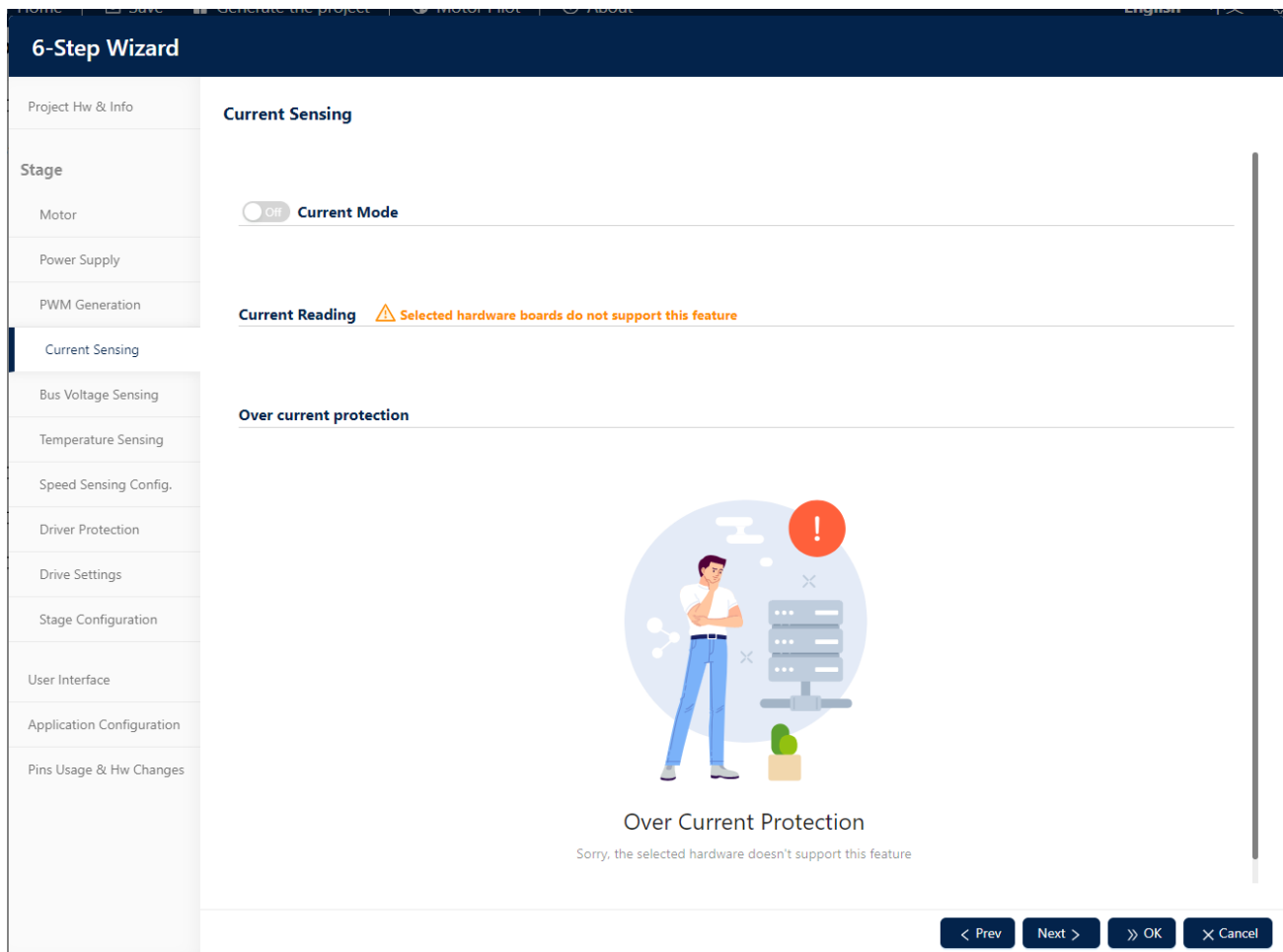


Figure 62: ST MC Workbench – 6 Step – Current Reading not available

3.7.8.8 Over Current Protection

From the Current Sensing step, the user can also enable and configure the Over Current Protection (OCP) with external comparator (Cf. Figure 63: ST MC Workbench – Current Sensing – Over Current Protection) or internal comparator (Cf. Figure 64: ST MC Workbench – Current Sensing – Over Current Protection Internal Comparators); the settings depend on the available OCP mode for the selected hardware. A schema will show the hardware connections between signals and MCU.

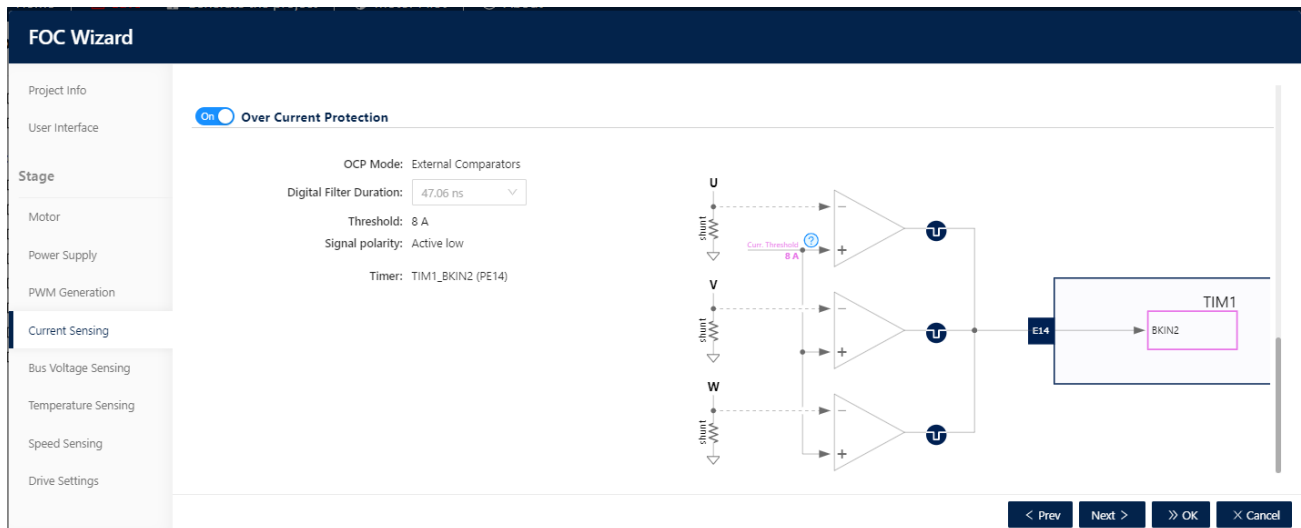


Figure 63: ST MC Workbench – Current Sensing – Over Current Protection External Comparators

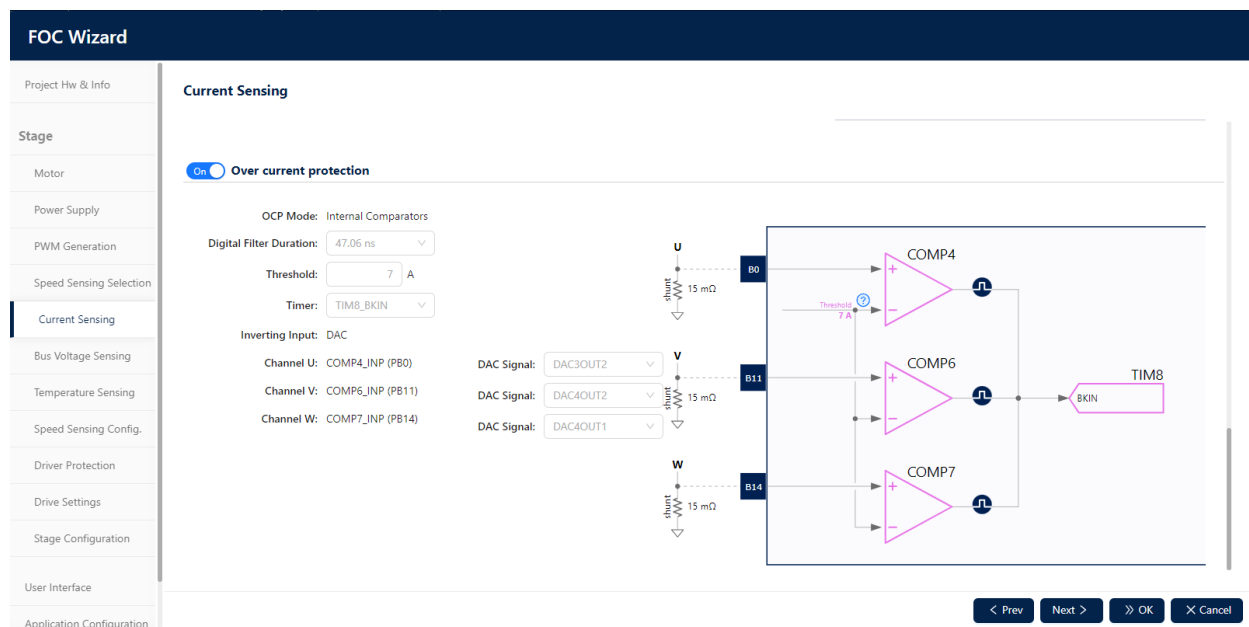


Figure 64: ST MC Workbench – Current Sensing – Over Current Protection Internal Comparators

3.7.9 Bus Voltage Sensing

In the Bus Voltage Sensing step, the user can configure the relevant parameters to sense the bus voltage (if available for the hardware specified) and the relative Over Voltage and/or Under Voltage protection.

FOC Wizard

Project Info

User Interface

Stage

Motor

Power Supply

PWM Generation

Current Sensing

Bus Voltage Sensing

Temperature Sensing

Speed Sensing

Drive Settings

On

Bus Voltage Sensing

MCU pin mapping

ADC: ADC1_IN7 (PC1)

Sampling time: 1 μ s (47.5 adc clk)

Bus Voltage: 325 Vdc

2.57 V

C1

ADC1

IN7

Protection

On

Over voltage

Threshold: 390 V (325 - 400 V)

Recovery action: Disable PWM generation

On

Under voltage

Threshold: 125 V (125 - 325 V)

< Prev

Next >

>> OK

✕ Cancel

Figure 65: ST MC Workbench – Bus Voltage Sensing

3.7.10 Temperature Sensing

In the Temperature Sensing step, the user can configure some useful parameters relevant to sense the temperature (if available for the hardware specified) and the relative Over Temperature protection.

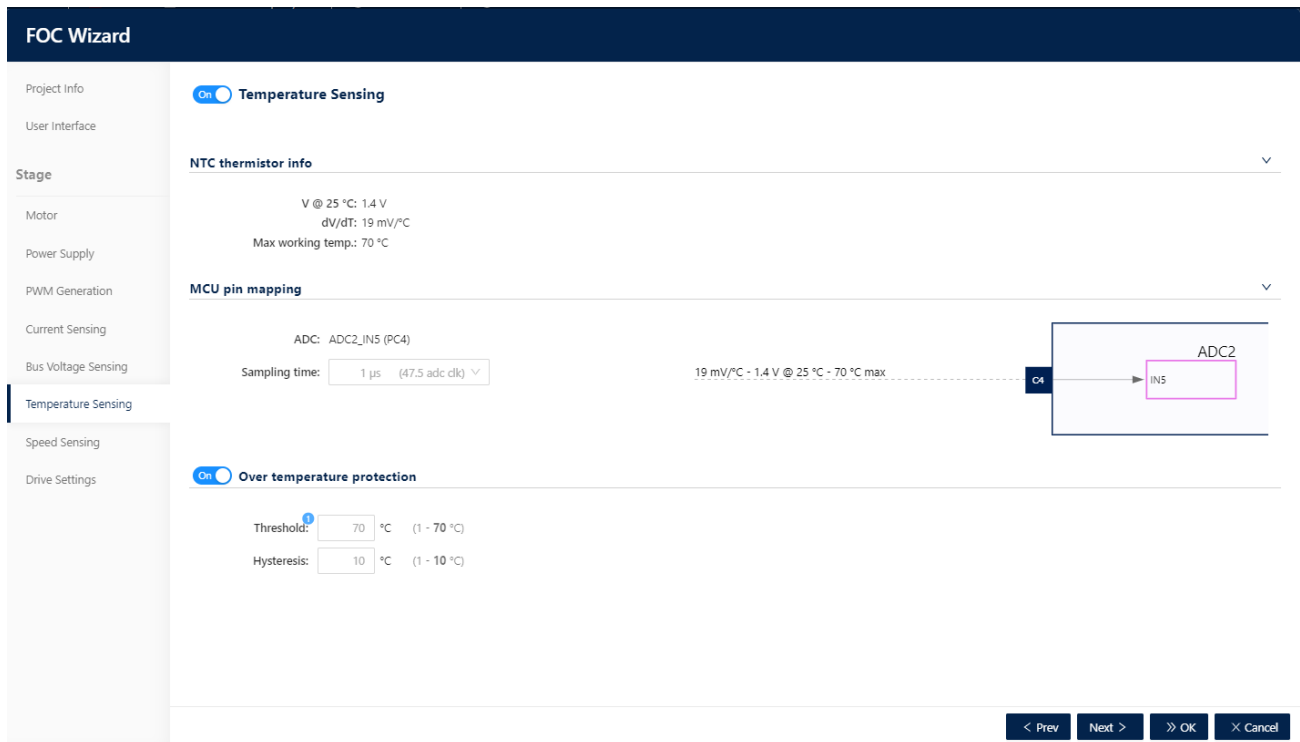


Figure 66: ST MC Workbench – Temperature Sensing

3.7.11 Speed Sensing Config

In the Speed Sensing step, the user can configure the main parameters in relation to the speed sensing mode selected in the previews Speed Sensing Selection step (Speed Sensing Selection ref paragraph 3.7.7); there are different speed sensing modes that are available for the Auxiliary Sensor and, for 6 Step Algorithm projects, also for the Main sensor (to change the Main Sensor Mode in FOC algorithm projects the user has to move to the previews Speed Sensing Selection Step). The available modes depend on the selected algorithm. Auxiliary sensor is optional, disabled by default.

3.7.11.1 FOC Observer + PLL (Sensorless)

This is the default speed sensing mode for the main sensor; it is always available.

FOC Wizard

Project Info | User Interface | Stage | Motor | Power Supply | PWM Generation | Current Sensing | Bus Voltage Sensing | Temperature Sensing | **Speed Sensing** | Drive Settings

Main Sensor | Auxiliary Sensor | Sensorless start-up parameters

Speed Sensor Mode:

Max Num. Errors before fault:

Observer + PLL (Sensorless)

Variance Threshold: %

Average speed depth for speed loop:

Average speed depth for observer equations:

B-emf consistency tolerance: %

B-emf consistency gain: %

☒ Sensorless speed feedback

Observer and PLL

☒ Auto Calculate

G1: -24453

G2: 24351

P: 665 / 16384 0.04059

I: 29 / 65536 0.00044

< Prev Next > >> OK X Cancel

Figure 67: ST MC Workbench – Speed Sensing – Observer PLL

3.7.11.2 FOC Observer + Cordic (Sensorless)

This speed sensor mode is always available.

FOC Wizard

Project Info | User Interface | Stage | Motor | Power Supply | PWM Generation | Current Sensing | Bus Voltage Sensing | Temperature Sensing | **Speed Sensing** | Drive Settings

Main Sensor | Auxiliary Sensor | Sensorless start-up parameters

Speed Sensor Mode:

Max Num. Errors before fault:

Observer + Cordic (Sensorless)

Variance Threshold: %

Average speed FIFO depth for speed loop:

Average speed FIFO depth for observer equations:

B-emf consistency tolerance: %

B-emf consistency gain: %

B-emf quality factor:

Maximum application acceleration: rpm/s

☒ Sensorless speed feedback

Observer

☒ Auto Calculate

G1: -24453

G2: 24351

< Prev Next > >> OK X Cancel

Figure 68: ST MC Workbench – Speed Sensing – Observer Cordic

3.7.11.3 FOC High Sensitivity Observer (HSO)

This speed sensor mode is available only for FOC driving mode, given that the hardware selected supports the HSO.

The screenshot shows the 'FOC Wizard' interface with a sidebar on the left containing the following stages: Project Hw & Info, Stage, Motor, Power Supply, PWM Generation, Speed Sensing Selection, Current Sensing, Bus Voltage Sensing, Temperature Sensing, Speed Sensing Config. (highlighted), Driver Protection, Drive Settings, Stage Configuration, User Interface, Application Configuration, and Pins Usage & Hw Changes. The main area is titled 'Main Sensor' and 'Sensorless start-up parameters'. It shows 'Speed Sensor Mode: High Sensitivity Observer (Sensorless)' with an information icon. Below this is the 'HSO' section with a 'Reset to default' button and the following parameters: Current Scale (29 A), Voltage Scale (167 V), Frequency Scale (367 Hz), Ksample delay (0.2), Angle compensation factor (2.67), Bus Protection (unchecked), Regeneration high limit (15.98 V), Regeneration low limit (14.21 V), Acceleration high limit (10 V), Acceleration low limit (7.5 V), and Soft. over current limit (6 A). The 'MCU pin mapping' section shows: ADC: ADC3 / ADC4, Phase Voltage U: ADC4_IN7 (PD10), Phase Voltage V: ADC4_IN16 (PE12), and Phase Voltage W: ADC3_IN10 (PD13). At the bottom right are buttons for '< Prev', 'Next >', '>> OK', and 'X Cancel'.

Parameter	Value	Unit
Current Scale	29	A
Voltage Scale	167	V
Frequency Scale	367	Hz
Ksample delay	0.2	
Angle compensation factor	2.67	
Regeneration high limit	15.98	V
Regeneration low limit	14.21	V
Acceleration high limit	10	V
Acceleration low limit	7.5	V
Soft. over current limit	6	A

Parameter	Value
ADC	ADC3 / ADC4
Phase Voltage U	ADC4_IN7 (PD10)
Phase Voltage V	ADC4_IN16 (PE12)
Phase Voltage W	ADC3_IN10 (PD13)

Figure 69: ST MC Workbench – Speed Sensing – High Sensitivity Observer (Sensorless)

3.7.11.4 FOC Quadrature Encoder

In order for this mode to be selectable, the motor needs to have the Quadrature Encoder sensor enabled and the feature must be supported by the hardware.

FOC Wizard

Project Info

User Interface

Stage

Motor

Power Supply

PWM Generation

Current Sensing

Bus Voltage Sensing

Temperature Sensing

Speed Sensing

Drive Settings

Main Sensor

Auxiliary Sensor

Speed Sensor Mode: Quadrature Encoder

Max Num. Errors before fault: 3

Quadrature Encoder

Average speed FIFO depth: 16

Input capture filter duration: 0.7 μ sec

Pulse per mechanical revolution: 400

☐ Reverse counting direction
 ☐ Use encoder alignment

Start-up parameters

Duration: 700 ms

Alignment electrical angle: 90 deg

Final current ramp value: 2.95 A

MCU pin mapping

Timer: TIM2

Encoder A: TIM2_CH1 (PA0)

Encoder B: TIM2_CH2 (PD4)

< Prev

Next >

>> OK

× Cancel

Figure 70: ST MC Workbench – Speed Sensing – Quadrature Encoder

3.7.11.5 Hall Sensor

In order for this mode to be selectable, the motor needs to have the Hall Effect sensor enabled and the feature must be supported by the hardware. This speed sensor mode is compatible with both FOC and 6 Step algorithm.

FOC Wizard

Project Hw & Info

Stage

Motor

Power Supply

PWM Generation

Speed Sensing Selection

Current Sensing

Bus Voltage Sensing

Temperature Sensing

Speed Sensing Config.

Driver Protection

Drive Settings

Stage Configuration

User Interface

Application Configuration

Pins Usage & Hw Changes

Main Sensor

Auxiliary Sensor

Speed Sensor Mode: ¹ Hall Sensor

Max Num. Errors before fault:

Hall Sensor

Average speed FIFO depth:

Input capture filter duration: μ sec

Sensors displacement: 120 deg

Placement electrical angle: 300 deg

☐ Disable electrical angle interpolation

MCU pin mapping

Timer: TIM2

Hall sensor H1: TIM2_CH1 (PA15)

Hall sensor H2: TIM2_CH2 (PB3)

Hall sensor H3: TIM2_CH3 (PB10)

< Prev

Next >

>> OK

✕ Cancel

Figure 71: ST MC Workbench – Speed Sensing with 6 step algo – Hall Sensor

3.7.11.6FOC Sensorless Start-up parameters

The settings for this section are available only if the main sensor selected is sensor less (Observer + PLL or Observer + Cordic).

FOC Wizard

Project Info

User Interface

Stage

Motor

Power Supply

PWM Generation

Current Sensing

Bus Voltage Sensing

Temperature Sensing

Speed Sensing

Drive Settings

Main Sensor

Auxiliary Sensor

Sensorless start-up parameters

Start-up profile

Initial electrical angle: 0 deg

On-the-fly start-up: Off

	Duration (ms)	Speed target (rpm)	Current target (A)
Phase 1:	1000	0	2.95
Phase 2:	3996	1998	2.95
Phase 3:	0	1998	2.95
Phase 4:	0	1998	2.95
Phase 5:	0	1998	2.95

Execute sensor-less algorithm starting from phase: 2

Start-up exit condition

Start-up speed threshold: 1798 rpm

Consecutive correct measures: 2

Estimated limits for the speed band tolerance

Lower limit: 93.75 %

Upper limit: 106.25 %

On Rev-up to FOC switch-over

Duration: 25 ms

Speed target (rpm)

Current target (A)

Speed target

Current target

< Prev

Next >

>> OK

✕ Cancel

Figure 72: ST MC Workbench – Speed Sensing – Sensorless start-up parameters

3.7.11.7FOC HSO Startup parameters

The following image shows the startup parameters available for HSO Speed Sensing mode.

FOC Wizard

Project Hw & Info

Stage

Motor

Power Supply

PWM Generation

Speed Sensing Selection

Current Sensing

Bus Voltage Sensing

Temperature Sensing

Speed Sensing Config.

Driver Protection

Drive Settings

Stage Configuration

User Interface

Application Configuration

Main Sensor

Sensorless start-up parameters

HSO start-up parameters

☒

RS DC Estimation Alignment

☒

PolPulse

Pulse current goal:

19.79

A

Pulse duration:

1

PWM period

Decay time:

1

PWM period

< Prev

Next >

>> OK

✕ Cancel

Figure 73: ST MC Workbench – Speed Sensing – HSO Sensorless start-up parameters

3.7.11.8 6 Step Sensorless ADC

Sensorless – ADC is a specific Speed Sensing mode for 6 Step algorithm (Cf. Figure 74: ST MC Workbench – 6 Step – Sensorless ADC).

6-Step Wizard

Project Hw & Info

Stage

Motor

Power Supply

PWM Generation

Current Sensing

Bus Voltage Sensing

Temperature Sensing

Speed Sensing Config.

Driver Protection

Drive Settings

Stage Configuration

User Interface

Application Configuration

Pins Usage & Hw Changes

Main Sensor

Sensorless start-up parameters

Speed Sensor Mode: Sensorless - ADC

Max Num. Errors before fault: 3

Sensorless - ADC

Average speed FIFO depth: 8

Bemf divider: 5.5

Threshold with high side recirculation: 100 % Bus Voltage

Computed threshold with high side recirculation: 2.27 V

Threshold with low side recirculation: 1 % Bus Voltage

PWM off Sampling point: 90 % of PWM cycle

PWM On Sensing: On

Threshold with PWM on: 50 % Bus Voltage

Computed threshold with PWM on: 1.32 V

PWM on Sampling point: 25 % of PWM cycle

On sensing enabling: 70 % of PWM cycle

On sensing enabling hysteresis: 5 % of PWM cycle

Falling BEMF zero crossing to step change delay: 30 electrical degrees

Rising BEMF zero crossing to step change delay: 30 electrical degrees

Max BEMF detection errors: 3

Delay computation step choice: Two Previous

Timer: TIM2

MCU pin mapping

Bemf Phase U: ADC2_IN6 (PC0)

Bemf Phase V: ADC1_IN9 (PC3)

Bemf Phase W: ADC2_IN7 (PC1)

Bemf divider: GPIO_Output (PC9)

Sampling time: 1 μ s (47.5 adc clk)

Number of ADC AWD samples: None

< Prev

Next >

>> OK

× Cancel

Figure 74: ST MC Workbench – 6 Step – Sensorless ADC

3.7.11.9 6 Step Sensorless Start-up parameters

6 Step algorithm has specific “Sensorless Start-up parameters” settings, available if Speed Sensing mode selected is Sensorless-ADC (Figure 75: ST MC Workbench – 6 Step – Sensorless – Start-up parameters). The Initial Position Detection is available only when current reading is enabled (6 Step Current Reading). Figure 61: ST MC Workbench – 6 Step – Current Reading).

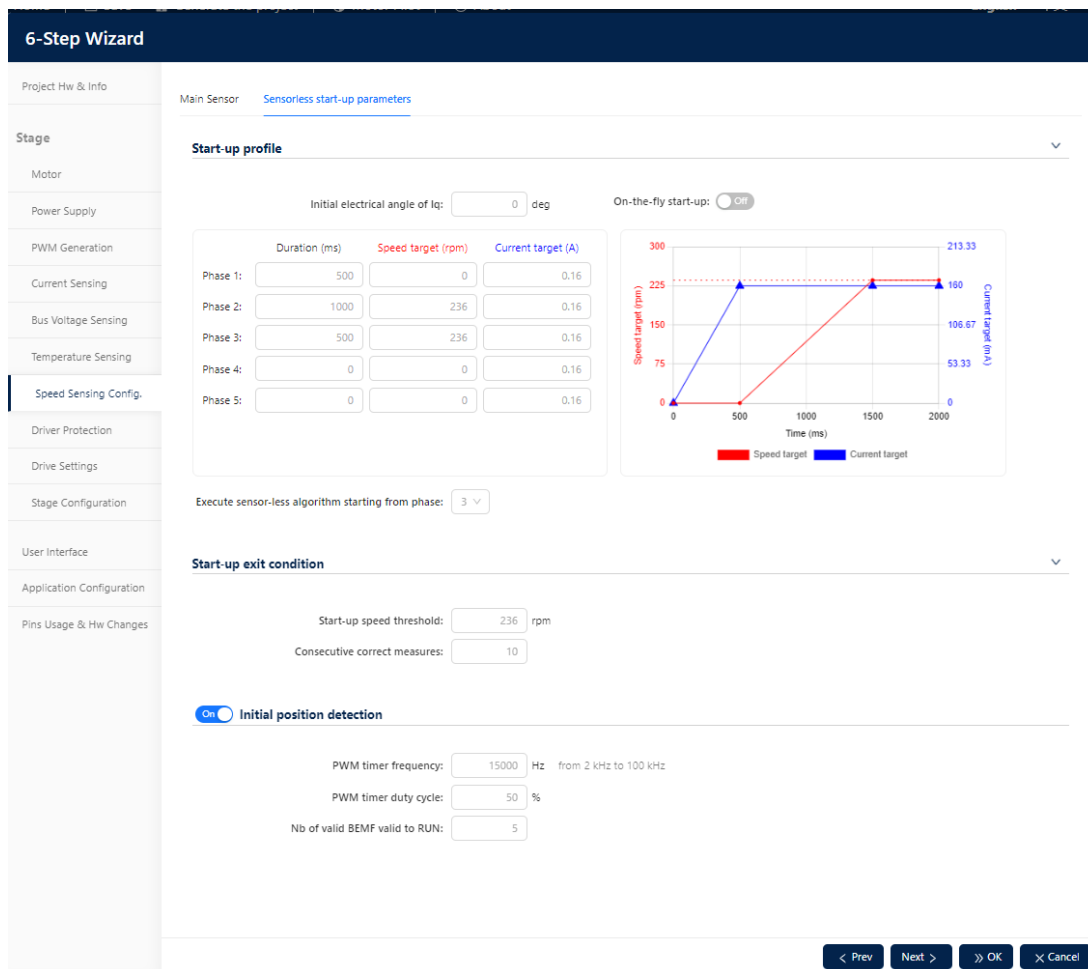


Figure 75: ST MC Workbench – 6 Step – Sensorless – Start-up parameters

3.7.12 Driver Protection

If available for the selected hardware, in this step the user can configure the driver protection parameters.

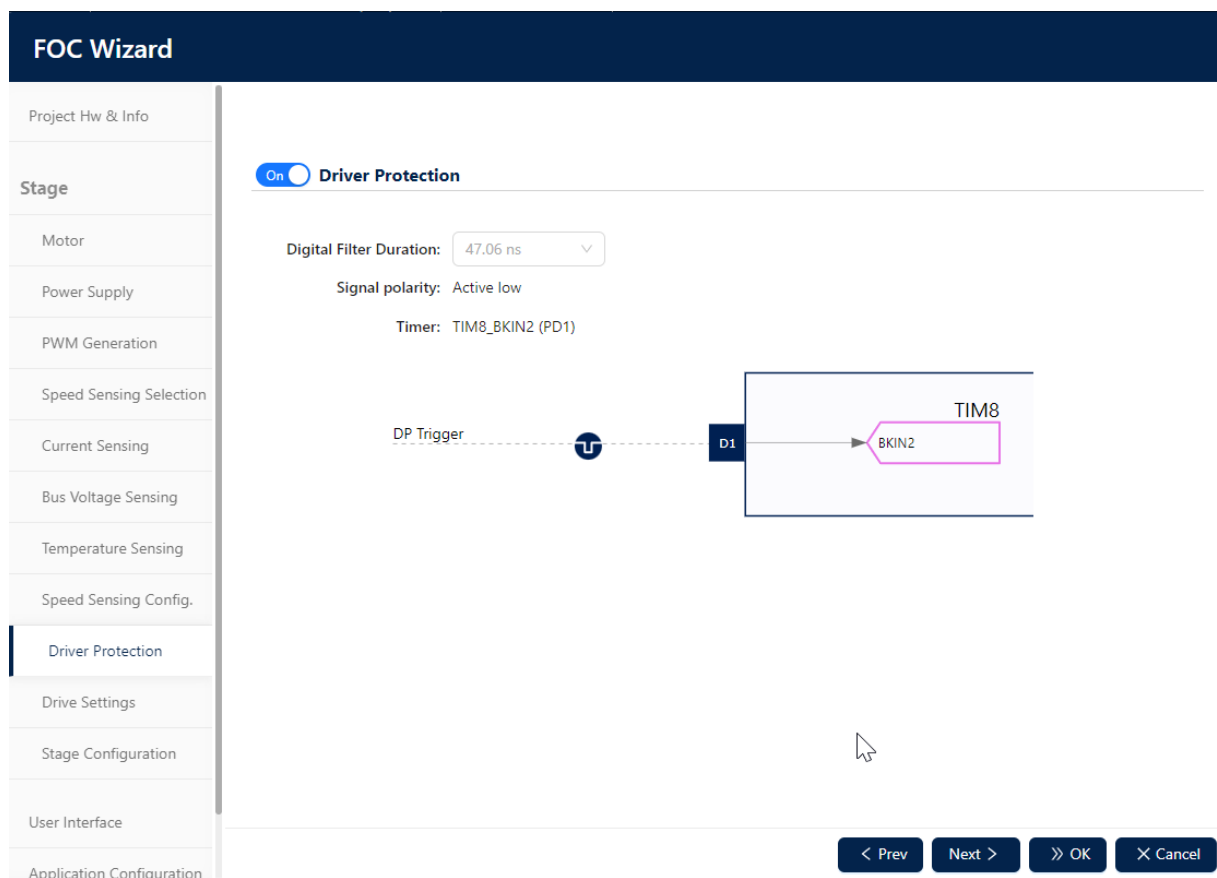


Figure 76: ST MC Workbench –Driver Protection

3.7.13 Drive Settings

In the Drive Settings step, the user can configure the Current Regulator as one of the following three control mode:

3.7.13.1 FOC Speed control

This drive settings mode is always available.

The screenshot displays the 'FOC Wizard' interface, specifically the 'Drive Settings' stage. The left sidebar lists various configuration stages, with 'Drive Settings' selected. The main area is titled 'Current regulator' and shows the 'Speed control' mode selected. Key settings include an execution time of 63 µs, a target speed of 1798 rpm, and an execution rate of 1 ms. The 'Auto Calculate Kp & Ki' option is checked. Below these settings, three control loops are visualized: the 'Speed regulator' (a PI controller with P=11 and I=0.0261), the 'Torque (Iq) current regulator' (a PI controller with P=1.47 and I=0.0147), and the 'Flux (Id) current regulator' (a PI controller with P=0.735 and I=0.0147). The 'Flux weakening regulator' is currently turned off. At the bottom, navigation buttons for '< Prev', 'Next >', '>> OK', and 'X Cancel' are visible.

Figure 77: ST MC Workbench – Drive Settings – Speed Control

3.7.13.2 FOC Torque control

This drive settings mode is always available.

FOC Wizard

Project Info

User Interface

Stage

- Motor
- Power Supply
- PWM Generation
- Current Sensing
- Bus Voltage Sensing
- Temperature Sensing
- Speed Sensing
- Drive Settings**

Current regulator

Execution time: 63 μ s

Control mode: Torque control

☐ Use MTPA

☐ Feed forward

☒ Auto Calculate Torque/Flux Kp & Ki

Cut-off frequency: 3000 rad/s

Torque (Iq) current regulator

Target: 0 A

P: 3012 / 2048 1.4707

I: 241 / 16384 0.01471

Flux (Id) current regulator

Target: 0 A

P: 1506 / 2048 0.73535

I: 241 / 16384 0.01471

Torque (Iq) current regulator block diagram:

0 A $\rightarrow$ Σ $\rightarrow$ P (1.47) $\rightarrow$ I (0.0147) $\rightarrow$ Σ

Flux (Id) current regulator block diagram:

0 A $\rightarrow$ Σ $\rightarrow$ P (0.735) $\rightarrow$ I (0.0147) $\rightarrow$ Σ

< Prev Next > >> OK X Cancel

Figure 78: ST MC Workbench – Drive Settings – Torque Control

3.7.13.3 FOC Position Control

This drive settings mode is available only if Quadrature Encoder is selected as main Speed Sensing sensor.

Project Info

User Interface

Stage

Motor

Power Supply

PWM Generation

Current Sensing

Bus Voltage Sensing

Temperature Sensing

Speed Sensing

Drive Settings

Current regulator

Execution time: 63 μ s

Control mode: Position control

☐ Use MTPA

☐ Feed forward

☒ Auto Calculate Torque/Flux Kp & Ki

Cut-off frequency: 3000 rad/s

Torque (Iq) current regulator

Flux (Id) current regulator

Position regulator

Execution rate: 1 ms

P: 10000 / 1024 9.76563

I: 1000 / 32768 0.03052

D: 1000 / 16 62.5

Flux weakening regulator

Off

Position regulator

Torque (Iq) current regulator

Flux (Id) current regulator

< Prev

Next >

>> OK

× Cancel

Figure 79: ST MC Workbench – Drive Settings – Position Control

3.7.13.4 FOC Speed control (for HSO)

This Drive setting is available in HSO Speed Sensing mode only.

Project Hw & Info

Stage

Motor

Power Supply

PWM Generation

Speed Sensing Selection

Current Sensing

Bus Voltage Sensing

Temperature Sensing

Speed Sensing Config.

Driver Protection

Drive Settings

Stage Configuration

User Interface

Application Configuration

Pins Usage & Hw Changes

FOC Wizard

Current regulator

Execution time: 33 μ s

Control mode: Speed control

(Id), (Iq) current regulator

P: 12.70094 V/A

I: 5000.00067 rad/s

Speed regulator

Execution rate: 1 ms

Target speed: 524 rpm

P: 0.5 A/Hz

I: 0 A/rad

Speed regulator

524 rpm $\rightarrow$ Σ $\rightarrow$ P 0.5 $\rightarrow$ I 0 $\rightarrow$ Σ

(Id), (Iq) current regulator

Σ $\rightarrow$ P 12.7 $\rightarrow$ I 5000 $\rightarrow$ Σ

< Prev

Next >

>> OK

✕ Cancel

Figure 80: ST MC Workbench – Drive Settings – FOC Speed Control (for HSO)

3.7.13.5 FOC Torque control (for HSO)

This Drive setting is available only in HSO Speed Sensing mode only.

FOC Wizard

Project Hw & Info

Stage

Motor

Power Supply

PWM Generation

Speed Sensing Selection

Current Sensing

Bus Voltage Sensing

Temperature Sensing

Speed Sensing Config.

Driver Protection

Drive Settings

Stage Configuration

User Interface

Application Configuration

Pins Usage & Hw Changes

Current regulator

Execution time: 33 μ s

Control mode: Torque control

Torque (Iq) current regulator

Flux (Id) current regulator

Target: 0 A
P: 6.29297 V/A
I: 1.04883 rad/s

Target: 0 A
P: 6.29297 V/A
I: 1.04883 rad/s

Torque (Iq) current regulator

Flux (Id) current regulator

0 A

0 A

Σ

P
6.29

I
1.05

Σ

Σ

P
6.29

I
1.05

Σ

Σ

P
6.29

I
1.05

Σ

Σ

P
6.29

I
1.05

Σ

< Prev

Next >

>> OK

✕ Cancel

Figure 81: ST MC Workbench – Drive Settings – FOC Torque Control (for HSO)

3.7.13.6 6 Step – Drive Setting

This step is available only on 6 Step projects. It is the Drive Setting step for this configuration.

The screenshot shows the '6-Step Wizard' interface in the ST MC Workbench. The left sidebar lists the configuration steps: Project Hw & Info, Stage, Motor, Power Supply, PWM Generation, Current Sensing, Bus Voltage Sensing, Temperature Sensing, Speed Sensing Config., Driver Protection, Drive Settings (selected), Stage Configuration, User Interface, Application Configuration, and Pins Usage & Hw Changes. The main area is titled '6-Step Wizard' and contains the following sections:

- Debug Settings:** Includes a checkbox for 'Open Loop'.
- Speed regulator:** Includes fields for 'Execution rate' (1 ms) and 'Target speed' (524 rpm). A checkbox 'Auto Calculate Kp & Ki' is checked, showing calculated values: P: 2758 / 8192, 0.33667 and I: 18 / 16384, 0.0011.
- Demagnetization management:** Includes fields for 'Revup demag. time' (20 % of step), 'Run demag. time' (20 % of step), 'Speed threshold' (786 rpm), and 'Minimum demag. time' (3 PWM cycles).
- Speed regulator block diagram:** A diagram showing a feedback loop. A '524 rpm' input is compared with a feedback signal at a summing junction (Σ). The error signal is processed by a 'P' block (0.337) and an 'I' block (0.0011), which are then summed at another junction (Σ) to produce the output.

Navigation buttons at the bottom right include '< Prev', 'Next >', '>> OK', and 'X Cancel'.

Figure 82: ST MC Workbench – Drive Settings – 6 Step

3.7.14 Stage Configuration

This step contains extra configuration parameters and a summary about protections available for the current stage.

FOC Wizard

Bus Voltage Sensing

Temperature Sensing

Speed Sensing Config.

Driver Protection

Drive Settings

Stage Configuration

Stage 2

Motor

Power Supply

PWM Generation

Speed Sensing Selection

Current Sensing

Bus Voltage Sensing

Temperature Sensing

Speed Sensing Config.

Driver Protection

Drive Settings

Stage Configuration

User Interface

Application Configuration

Pins Usage & Hw Changes

Debug Settings

☐ Open Loop

☒ Potentiometer

MCU pin mapping

ADC: ADC1_IN9 (PC3)

Sampling time: 1 μ s (47.5 adc clk)

LEVEL

ADC1

IN9

Electronic Speed Control ⚠ Selected hardware boards do not support this feature

Protections summary

☒ Over current protection (Enabled)
To change Over current protection, refer to Current Sensing step (click [here](#) to go to such step).

☐ Driver Protection (Disabled)
To change Driver protection, refer to Driver Protection step (click [here](#) to go to such step).

☒ Over temperature protection (Enabled)
To change Over temperature protection, refer to Temperature Sensing step (click [here](#) to go to such step).

☒ Bus Voltage Sensing / Over voltage (Enabled)
To change Over voltage, refer to Bus voltage Sensing step (click [here](#) to go to such step).

☒ Bus Voltage Sensing / Under voltage (Enabled)

< Prev

Next >

>> OK

✕ Cancel

Figure 83: ST MC Workbench – Stage Configuration

3.7.15 User Interface

This is a common step (available for all the stages) that lets the user configure “Start and Stop button” and the “Motor Control Protocol (MCP)”, given that these features are available for the selected hardware.

The screenshot displays the 'FOC Wizard' interface. On the left is a vertical sidebar with a list of configuration stages: Bus Voltage Sensing, Temperature Sensing, Speed Sensing Config., Driver Protection, Drive Settings, Stage Configuration, Stage 2, Motor, Power Supply, PWM Generation, Speed Sensing Selection, Current Sensing, Bus Voltage Sensing, Temperature Sensing, and Speed Sensing Config. The 'Stage 2' section is currently active. The main area of the wizard is divided into two primary sections. The first section, 'Start/Stop on Control board', is enabled (indicated by a blue 'On' toggle) and shows settings for Pin: PC13, Trigger edge: Rising Edge, and Internal pull up/down: No Pull up - No Pull down. The second section, 'Motor Control Protocol', is also enabled (blue 'On' toggle) and shows settings for Peripheral: USART1_TX (PA9) - USART1_RX (PA10), Baudrate: 1843200, Data Log (checked), Max Signals to plot: 10, and Buffer Size: 2048. Below this is a 'Shared Buffer Size' section with a Buffer Size of 128. At the bottom right of the main area are four navigation buttons: '< Prev', 'Next >', '>> OK', and 'X Cancel'.

Figure 84: ST MC Workbench – User Interface

The user can only edit configurations that are available for the specific selected hardware. In example visible in the above image, for instance, the start/stop is available on PC13 pin only. If the selected hardware allows more possibilities a combo will be available to allow user to choose a different pin. The Start/Stop feature can be disabled if the user doesn't want to use it.

3.7.16 Application Configuration

This step is a common step that allows generic configurations unrelated to any specific stage.

The screenshot shows the 'FOC Wizard' interface with the 'Application Configuration' stage selected in the left sidebar. The main area is divided into two sections: 'FreeRTOS' and 'DAC Debug'. The 'FreeRTOS' section has a toggle switch set to 'On' and a 'Timer' dropdown menu set to 'TIM6'. Below this is a checkbox labeled 'Allow MC IRQ to use FreeRTOS functions' with a blue information icon. The 'DAC Debug' section has a toggle switch set to 'Off' and three checkboxes: 'Reduce the number of registers accessible with the MC Pilot to fit in 32 KB Flash', 'MCU Load measurement', and 'Motor Profiler'. The 'Motor Profiler' checkbox has a red question mark icon. At the bottom right, there are four buttons: '< Prev', 'Next >', '>> OK', and 'X Cancel'.

FOC Wizard

Stage 2

Motor

Power Supply

PWM Generation

Speed Sensing Selection

Current Sensing

Bus Voltage Sensing

Temperature Sensing

Speed Sensing Config.

Driver Protection

Drive Settings

Stage Configuration

User Interface

Application Configuration

Pins Usage & Hw Changes

FreeRTOS

On

Timer: TIM6

☐ Allow MC IRQ to use FreeRTOS functions

DAC Debug

Off

☐ Reduce the number of registers accessible with the MC Pilot to fit in 32 KB Flash

☐ MCU Load measurement

☐ Motor Profiler

< Prev Next > >> OK X Cancel

Figure 85: ST MC Workbench – Application Configuration

3.7.17 Pin usage & Hardware Changes

This step shows a summary about all the pins used in the project. Any detected pin conflict is alerted by a red triangle . The conflict is also alerted in the correlated steps.

FOC Wizard

Drive Settings

Stage Configuration

Stage 2

Motor

Power Supply

PWM Generation

Speed Sensing Selection

Current Sensing

Bus Voltage Sensing

Temperature Sensing

Speed Sensing Config.

Driver Protection

Drive Settings

Stage Configuration

User Interface

Application Configuration

Pins Usage

Hardware Changes

Conflicts	Stage	IP	Channel	Pin	Signal name
	1	ADC1	IN3	PA2	OPAMP1_OUT1
	1	COMP2	INP	PA3	CURRENT_SHUNT_P
	1	ADC1	IN9	PC3	POTENTIOMETER_LEVEL
		DAC1	OUT1	PA4	DAC_EMUL_CH1
	2	ADC1	IN7	PC1	VBUS
	2	TIM8	CH1	PA15	PWM_CHU_H
	2	TIM8	CH2	PB8-BOOT0	PWM_CHV_H
	2	TIM8	CH3	PB9	PWM_CHW_H
	2	GPIO	Output	PB7	PWM_EN
	2	OPAMP2	VINP	PA7	CURRENT_SHUNT_P
	2	OPAMP2	VINM	PA5	CURRENT_AMPLN
	2	OPAMP2	VOUT	PA6	CURRENT_AMPL
	2	ADC2	IN3	PA6	OPAMP2_OUT
	2	COMP2	INP	PA7	CURRENT_SHUNT_P
	2	ADC2	IN17	PA4	POTENTIOMETER_LEVEL
		DAC1	OUT1	PA4	DAC_EMUL_CH1

Pins Usage & Hw Changes

< Prev

Next >

>> OK

✕ Cancel

Figure 86: ST MC Workbench – Pin Usage

In the same step, the Hardware Changes section shows the hardware changes needed (if any) to make the project properly work with the user's configurations.

FOC Wizard

Stage

Motor

Power Supply

PWM Generation

Speed Sensing Selection

Current Sensing

Bus Voltage Sensing

Temperature Sensing

Speed Sensing Config.

Driver Protection

Drive Settings

Stage Configuration

User Interface

Application Configuration

Pins Usage & Hardware Changes

Pins Usage

Hardware Changes

Warning! Hardware changes must be applied to the following signals. Please, refer to the board diagram for details.

Pin	Signal name	Help
PA0	BUTTON_TRIGGER	Close SB21.
PA15(JTDI)	HALLSENSOR_H1	CN7.17. T_JTDI default feature signal. Remove SB9 for encoder/hall feature Enc. A/H1 signal
PB3(JTDO/TRACESW O)	HALLSENSOR_H2	CN10.31. T_SWO default feature signal. Remove SB15 for encoder/hall feature Enc. A/H1 signal

< Prev

Next >

>> OK

✕ Cancel

Figure 87: ST MC Workbench – Hardware Changes